Pre-Stall Load Redistribution on Swept Tailplane Models with Full-Span Controls

Andrea Rausa*, Alex Zanotti†, Giuseppe Gibertini‡, Alberto Guardone§, and Franco Auteri¶
Politecnico di Milano, Department of Aerospace Science and Technology, Milano, Italy, 20154

Local loading changes on a swept lifting surface with a deflected plain control can alter lift growth and pitching moment well before maximum lift. MONNALISA wind-tunnel measurements and Reynolds-averaged Navier–Stokes calculations are used to examine how fixed-surface and full-span control loads combine in this response. For the positive-sweep, zero-dihedral model with a 20° control deflection, the computed fixed surface continues gaining lift while both surface contributions fall below their preceding incidence trends. Across the tested fitting ranges, the fixed surface accounts for about 63% of the total lift departure; the control contributes more strongly to the moment departure about the fitted aerodynamic center. Pressure sections and regional integrations connect these different component contributions to stronger leading-edge suction, reduced hinge/aft loading, and longer outboard moment arms. Matched dihedral comparisons resolve incidence-dependent changes in lift growth and the force contributions to pitching moment. The forward-swept planform gives weaker aft losses and a different spanwise distribution of moment contributions. The detailed interpretation concerns the computed flow fields and loads; agreement with measured integrated coefficients is partial. The combined force and moment analysis identifies loading changes that continuing total lift growth alone would conceal.

Nomenclature

C_D, C_L	=	drag and lift coefficients
$C_{F,k}$	=	body-force coefficient, $k = x, y, z$
$C_M^A, C_M^O, C_M^{A_I}$	=	moments about aerodynamic center, origin, or common configuration-I axis
$C_p,\ \mathbf{C}_f, C_{f,c}$	=	pressure, vector skin-friction, and chordwise skin-friction coefficients
$D_{Q,j}$	=	departure from the preceding coefficient trend

*PhD, Department of Aerospace Science and Technology, Corresponding Author, andrea.rausa@polimi.it
†Associate Professor, Department of Aerospace Science and Technology
‡Associate Professor, Department of Aerospace Science and Technology
§Full Professor, Department of Aerospace Science and Technology
¶Full Professor, Department of Aerospace Science and Technology

F_k = body-force component, N

$\mathcal{I}_M$ = difference in moment increments between dihedral cases

M_∞ = freestream Mach number

Q = coefficient label, L or M

$Re_b, Re_{\bar{c}}$ = semispan and mean-chord Reynolds numbers

S = undeformed semiplanform reference area, m^2

T_∞ = freestream temperature, K

U_∞ = freestream speed, m/s

b = undeformed semispan, m

$c,\ \bar{c}$ = local and mean aerodynamic chords, m

f_{rev} = upper control surface area fraction with reverse flow

i, j, k = sample, surface-contribution, and force-component indices

$m_{Q,j}^{\mathrm{pre}}$ = preceding slope, per degree

p, p_∞ = surface and freestream pressures, Pa

q_∞ = freestream dynamic pressure, Pa

s = undeformed span coordinate, m

t = section thickness, m

$\mathbf{t}_c$ = downstream chordwise unit tangent

u_i, w_i = supplied sample uncertainty magnitude and linear-combination weight

$u_{\mathrm{ind}}, u_{\mathrm{L1}}$ = uncertainty sensitivities for coefficient departures

x, y, z = downstream, spanwise, and upward body coordinates, m

y^+ = nondimensional wall distance

x_A, x_O = aerodynamic-center and origin abscissae, m

$\alpha, \alpha_a, \alpha_b$ = incidence and interval endpoints, deg

Γ, δ = dihedral and control deflection, deg

$\Delta C_{Q,j}$ = actual coefficient increment

$\eta = s/b,\ \xi$ = span fraction and straightened local chord fraction

λ, Λ = quarter-chord sweep (deg) and aspect ratio

ρ_∞ = freestream density, kg/m^3

$\tau, \boldsymbol{\tau}_w$ = taper ratio and wall-shear vector (Pa)

I. Introduction

A deflected trailing-edge control changes both the amount of lift generated by a lifting surface and the location at which that load acts. Its response with incidence is therefore relevant to pitching moment as well as lift, even before the maximum-lift condition is reached. When local loading changes, total lift can continue increasing while its growth slows and the pitching response changes more abruptly. Interpreting this behavior requires relating the integrated coefficients to the loads on the fixed and moving parts of the surface.

Sectional plain-flap experiments establish the starting point for this problem. Sears and Liddell [1] measured lift, drag, pitching moment, and hinge moment for a 30%-chord plain flap on an NACA 0015 airfoil, including sealed and unsealed hinge gaps. Separation reduced flap effectiveness at large deflections, although additional deflection still increased lift within the tested range. Their comparison with a thinner symmetric section also showed that similar lift effectiveness could coexist with different hinge-moment behavior. Reduced flap effectiveness is thus established physics; neither the presence of separation nor a single integrated lift trend fully characterizes the loading response.

On a finite wing, flap-induced loading is also shaped by the spanwise flow and planform. Campbell, Blanks, and Leaver [2] investigated rectangular wings of small aspect ratio with full-span plain flaps of several chord fractions. Their force and pitching-moment measurements, compared with thin-wing theory and empirical thickness and boundary-layer allowances, related flap effectiveness and center of pressure to flap chord and aspect ratio. Those unswept, thinner, transition-tripped models differ from the present swept models, but they establish why the location of the additional load matters alongside its magnitude.

For swept wings, Harper and Maki [3] documented how spanwise changes in local lift response can produce pitching-moment nonlinearity below maximum lift. The first appearance of local stall need not coincide with the first integrated moment change, because the force change and its moment arm vary along the span. Their broad stall review provides context for examining load redistribution, rather than an equivalent test of the present deflected-control configurations.

Swept-flap experiments combining pressure, load, and flow-field measurements show that this spanwise organization extends to the flap flow itself. On a flapped NACA 0012 wing with adjustable sweep, Tewes et al. [4] found in the unactuated unswept configuration that the lift slope remained close to its thin-airfoil value although the flow over the deflected flap was separated, whereas the pitching moment increased as the separation location on the flap moved upstream. At 45° sweep and zero incidence, the unactuated wing with a large flap deflection produced about half

the lift of the unswept wing, and its pitching moment increased almost linearly with incidence instead of remaining constant. On a 30° swept NACA 0015 wing with a 30%-chord flap, Melton [5] reported baseline pressure plateaus on the deflected flap, a large recirculation region near midspan with spanwise velocity directed toward the tip, and a smaller recirculation region inboard. These lower-Reynolds-number studies primarily address active separation control and do not provide quantitative comparisons for the present models. Their unactuated results nevertheless show that flap separation can accompany continued lift growth while altering pitching moment, and that its extent varies along a swept span. They do not resolve how the fixed surface and flap separately contribute to those integrated changes.

Nonlinear sectional data provide one route for representing these responses in reduced-order models. Sivells and Neely [6] coupled such lift data to lifting-line theory and also calculated wing pitching moment from sectional moment data. Goitia and Llamas [7] coupled a vortex-lattice formulation to viscous sectional data, with effective-sweep and finite-wing corrections, to predict nonlinear wing and tail lift. Consequently, the ability of a reduced-order model to reproduce the response depends on how sectional nonlinearities and three-dimensional loading changes are represented.

The immediate basis for the present work is the MONNALISA database publication [8]. Its Sec. IV and Fig. 11 already document the baseline pre-stall lift nonlinearity, expansion of reverse flow on the control, associated pressure changes, and continued growth of total lift. Its computations with the DUST mid-fidelity solver of Tugnoli et al. [9] use sectional tables and corrections, while the retained potential-based panel and vortex-lattice predictions are smoother. The present loading analysis builds on these established observations.

The remaining loading question is how changes on the fixed surface and control combine to produce the joint lift and moment response. Continued fixed-surface lift growth, for example, does not establish that this contribution follows its preceding trend. Comparing actual increments with departures from that trend identifies which components supply the reduced total growth, while spatial pressure integrations show why the relative component contributions can differ for moment. This study develops that quantitative interpretation from the existing Reynolds-averaged Navier–Stokes (RANS) solutions, first assessing their integrated coefficients against the measured response and its supplied uncertainty. Changes in control deflection and geometry then test how broadly the reference loading pattern persists within the dataset.

The test articles are isolated tailplane-inspired wings with full-span, 30%-chord plain controls; no fuselage, wing wake, or complete-aircraft trim condition is represented. Dihedral is varied on the same planform, while a forward-swept planform provides the contrasting sweep direction. These comparisons define the scope of the analysis without establishing behavior for partial-span controls, other flap chords, or intermediate dihedral angles.

II. Methodology

A. Models and wind-tunnel setup

Wind-tunnel (WT) measurements establish the integrated lift and pitching-moment response, while RANS surface loads identify how the fixed surface and control contribute to it. To examine dihedral, the reference aft-swept planform (configuration I) is investigated at $\Gamma = 0^\circ$ and $\Gamma = 45^\circ$, with its in-plane geometry unchanged. A forward-swept planform (configuration II) provides a contrast in sweep direction and overall planform response. The labels I and II are used in the tables.

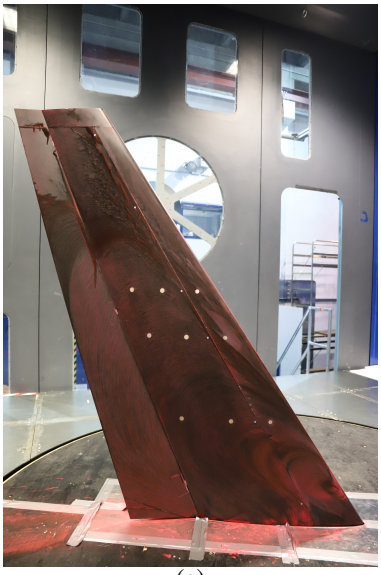
Both models use a symmetric section of constant profile along the undeformed semispan $b = 1.6$ m, with thickness ratio $t/c = 9\%$, where c is the local chord. Each model has a full-span plain control hinged at 70% chord. The reference surface S is the undeformed semiplanform area, and $\bar{c}$ denotes its mean aerodynamic chord (MAC). Table 1 gives the geometric parameters and coefficient references. The quarter-chord sweep is $\lambda = +30^\circ$ for configuration I and $\lambda = -15^\circ$ for configuration II; positive dihedral Γ raises the tip. Aspect ratio is defined for the corresponding full planform as $\Lambda = 2b^2/S$, and τ is the tip-to-root chord ratio. Configuration II also differs in taper, aspect ratio, MAC and chord-based Reynolds number, so the sweep contrast is interpreted through the overall planform response.

Table 1 Model geometry and coefficient reference quantities.

Property	I, $\Gamma = 0^\circ$	I, $\Gamma = 45^\circ$	II, $\Gamma = 0^\circ$
λ (deg)	30	30	-15
Λ	5.0	5.0	5.2
τ	0.4	0.4	0.6
S (m^2)	1.024	1.024	0.984604
$\bar{c}$ (m)	0.679184	0.679184	0.6282
Root chord (m, rounded)	0.914286	0.914286	0.76923
$Re_{\bar{c}}$	2.335×10^6	2.335×10^6	2.159×10^6
Numerical x_A (m)	0.642000	0.643532	-0.003618

The measurements were obtained in the low-turbulence section of the Galleria del Vento del Politecnico di Milano (GVPM), a closed-circuit facility. The test section is 4.00 m wide and 6.00 m long, with an original height of 3.84 m. For the present tests, installation of the false floor reduced the test-section height to 3.49 m; the undeformed model semispan is 40% of its width. A curved upstream inlet and gap beneath the false floor were designed to divert the incoming tunnel-floor boundary layer below the model installation. Boundary-layer development over the raised floor and around the support remains distinct from a numerical symmetry boundary. The installation photograph in Fig. 1 was taken at $\alpha = 25^\circ$, $\delta = 20^\circ$ and illustrates the mounting, rather than the pre-stall flow conditions.

The database setup description [8] gives freestream turbulence below 0.1%, measured using a turbulence sphere and hot-wire anemometry. Outside the wall layers, the inlet total-pressure variation was within $\pm 0.5\%$ of dynamic pressure



(a)

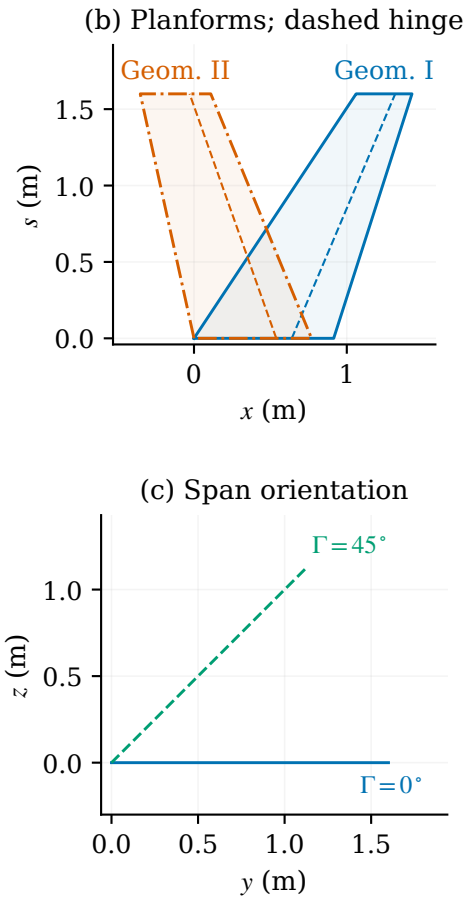


Fig. 1 Wind-tunnel installation and geometry: (a) mounted semispan model; (b) reference and forward-swept planforms; (c) zero- and positive-dihedral arrangements.

over most of the section and within ±1% overall; outlet measurements confirmed similar uniformity. The reported longitudinal static-pressure gradient is $\partial C_P/\partial x = -0.004$ m^{-1}. Loads were acquired through a RUAG 767 strain-gauge balance below the floor and an HBM QuantumX system; the reduction includes the balance-to-wind transformation and the database corrections. The nominal freestream Mach number and temperature are $M_\infty = 0.147$ and $T_\infty = 300$ K, with MAC Reynolds numbers given in Table 1. The common database operating point is $Re_b = 5.5 \times 10^6$; the MAC-based values are $Re_{\bar{c}} = Re_b \bar{c}/b$.

Recorded incidences are $\alpha = -25, -24, -22, \ldots, 22, 24, 25°$: the local pre-stall spacing is $\Delta\alpha = 2°$, with $\Delta\alpha = 1°$ endpoint steps. The comparisons use $\delta = 0, 10, 20, 30°$. Uncertainty bars are the supplied propagated measurement/reduction-chain magnitudes, including balance and transformation contributions. Each moment about the corresponding aerodynamic center retains its associated propagated uncertainty. The uncertainty sensitivities below use these supplied magnitudes.

B. Conventions and moment references

Positive incidence α raises the leading edge relative to the freestream, and positive control deflection δ moves the trailing edge downward relative to the fixed section. Lift is positive toward the upper surface and pitching moment is positive nose-up. The WT and numerical data are expressed in these common physical conventions. In body axes, x is downstream, y is spanwise, and z is upward. The axial force is the downstream body-axis component F_x, and the body-normal force is the upward component F_z. The latter is defined in the common body frame, rather than normal to a panel tilted by dihedral. The body-force components F_k, with $k = x, y, z$, are normalized by the freestream dynamic pressure $q_\infty = \rho_\infty U_\infty^2/2$ and area S, where ρ_∞ and U_∞ are freestream density and speed. Lift and drag coefficients follow from

$$C_L = C_{F,z}\cos\alpha - C_{F,x}\sin\alpha, \qquad C_D = C_{F,z}\sin\alpha + C_{F,x}\cos\alpha, \qquad C_{F,k} = F_k/(q_\infty S). \tag{1}$$

Pitching moment is normalized by $q_\infty S\bar{c}$. Body coordinates are expressed relative to the root leading edge.

Both experimental and numerical moments are referred to aerodynamic centers determined from the zero-deflection loading characteristics. Linear fits to moment and body-normal force over $|\alpha| \le 10^\circ$ define the positions, which remain fixed when control deflection varies [8]. The experimental fits use the supplied uncertainty weights; the numerical fits use the retained zero-deflection linear trends. For configuration I, the WT and RANS positions are 0.6430 and 0.6420 m at zero dihedral, and 0.6455 and 0.6435 m at $\Gamma = 45^\circ$: separations of about 1.0 and 2.0 mm. For configuration II they are -0.0492 and -0.00362 m, separated by 45.6 mm. The comparison coefficient C_M^A therefore uses the same reference construction, with each dataset's fitted position; it is not an imposed quarter-MAC moment.

The original numerical reference is $O = (0.228571, 0, 0)$ m, the root quarter-chord of Geom. I. For a numerical pole transported from abscissa x_0 to x_1,

$$C_M(x_1) = C_M(x_0) + \frac{x_1 - x_0}{\bar{c}} C_{F,z}. \tag{2}$$

The fitted abscissae are in Table 1. Matched dihedral diagnostics use the identical numerical axis $A_I = (0.642000, 0, 0)$ m in both configurations, denoted $C_M^{A_I}$. Geom. II uses its own MAC once for every moment contribution; all coefficients use the fixed areas in Table 1.

C. Numerical setup

The SU2 [10] calculations solve compressible ideal-gas RANS with Sutherland viscosity, the Spalart–Allmaras (SA) turbulence model [11], and the Bas–Cakmakcioglu (BCM) algebraic transition treatment [12]. The finite-volume mean-flow discretization uses median-dual vertex cells and Roe fluxes with second-order monotonic upstream-centered schemes for conservation laws (MUSCL) reconstruction [13], Green–Gauss gradients, and a Venkatakrishnan limiter [14] with coefficient 0.05. SA convection uses first-order scalar upwinding.

The model surfaces have no-slip, adiabatic boundaries, with freestream conditions at the far field and symmetry at the root. The symmetry plane omits the physical support and boundary-layer development over the raised-floor arrangement. Steady solutions use implicit Euler pseudo-time advancement and flexible generalized minimal residual (FGMRES) linear solves, with adaptive Courant–Friedrichs–Lewy (CFL) number. The production stopping criterion monitors lift and drag over 100 entries with a Cauchy tolerance of approximately 3×10^{-6}. Case-specific solution sensitivity is discussed where it affects the interpretation.

The surface grid resolves the leading edge, hinge junction, and trailing edge, where curvature and pressure changes require additional resolution. Wall-normal boundary-layer extrusion uses a nominal first-cell height of about 2×10^{-6} m and growth ratio 1.15, while volume refinement follows the wake at each incidence. The volume mesh is hex-dominant and octree-based [8]. Representative solutions have median nondimensional wall distance $y^+ = 0.19$–0.26, with local maxima about 1.1–1.3. A reference grid contains approximately 9.44 million points and 18.92 million volume elements.

The existing grid-refinement assessment uses four levels—coarse, nominal, fine, and very fine—for configuration I at $\delta = 35°$, $\Gamma = 0, 45°$, and $\alpha = 0, 18°$. Table 2 gives the point and element ranges. Figure 2 compares lift, drag, and pitching moment against the actual grid-point count on a logarithmic scale; open squares identify the nominal level, and lines connect the retained levels. Moments are about the numerical origin O, using the configuration-I area and MAC.

The nominal level was selected to balance spatial resolution with the demands of the extensive MONNALISA campaign across geometries, incidences, and control deflections. Its approximately 9.7–12.7 million points compare with 50.1–65.9 million on the finest level. Nominal-to-finest differences are 1.20–2.14% for lift and 1.13–2.49% for drag. At zero dihedral and $\alpha = 18°$, C_M^O changes from approximately -0.813 to -0.827, an absolute difference of 0.0134; the $\Gamma = 45°$ moment sequence is nonmonotonic. The assessment supports the nominal grid as a practical production compromise while showing finite, condition-dependent integrated-load sensitivity. It provides no discretization-error bound for local component loads. The comparison with measurements in Sec. III separately assesses reproduction of the principal pre-stall aerodynamic trends.

Table 2 Point and volume-element ranges, in millions, for the four-level refinement assessment.

Refinement	Points	Volume elements
Coarse	4.79–5.29	9.51–10.90
Nominal	9.67–12.74	19.06–24.07
Fine	22.33–26.56	39.93–45.70
Very fine	50.11–65.89	87.35–109.98

D. Load decomposition and trend departures

A surface can gain lift while contributing to a reduction in total lift growth. To distinguish these effects, the actual increment is the measured or computed coefficient change between two incidences. The expected increment is obtained

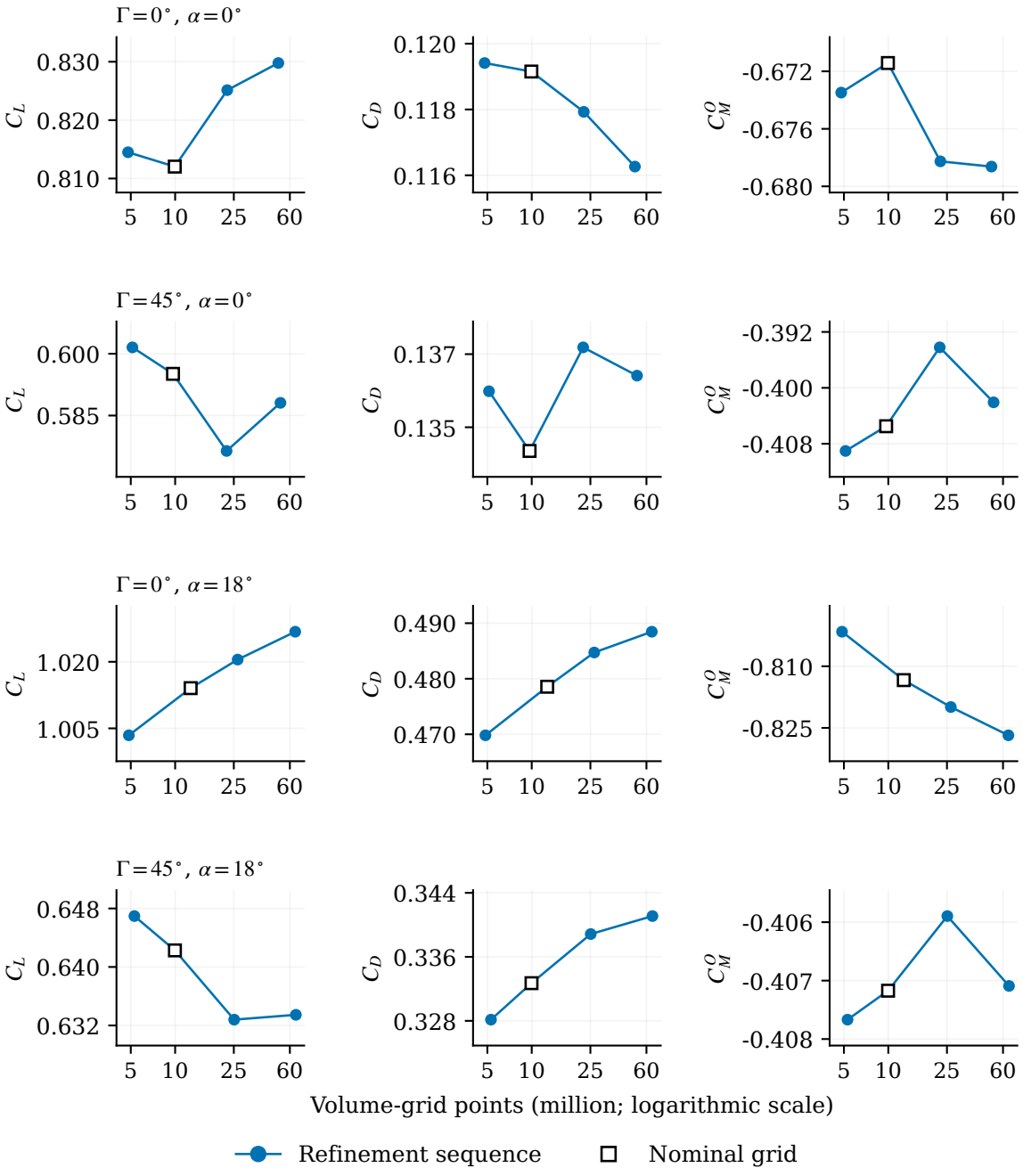


Fig. 2 Grid-refinement trends of lift, drag, and pitching moment for configuration I at $\delta = 35°$, two incidences, and two dihedral settings.

by linearly extrapolating the empirically fitted preceding slope over the same interval. Their difference, actual minus expected, is the departure. For the fixed surface of configuration I at zero dihedral and $\delta = 20°$, the actual increase over $\alpha = 0 \rightarrow 2°$ is +0.0593, while continuation of the preceding trend predicts +0.1285. The departure is −0.0692: the fixed surface still gains lift, but gains 0.0692 less than predicted by that trend. This is a reduction in growth, not an actual lift loss.

The analysis resolves these changes into additive surface loads. Solver surface groups define the fixed surface (upper/lower fixed panels), the control (upper/lower control panels, hinge junction, and blunt trailing edge), and tip. Pressure and viscous contributions are retained on both physical sides with the same global S and $\bar{c}$. Tip loads remain in every total and closure check. All displayed coefficients use their actual magnitude.

For coefficient label $Q = L$ or M and surface contribution j (fixed, control, tip, or total), the actual increment is the change between sampled incidences α_a and α_b. Its departure subtracts the change predicted by continuing the preceding slope $m^{\mathrm{pre}}_{Q,j}$, measured per degree,

$$\Delta C_{Q,j} = C_{Q,j}(\alpha_b) - C_{Q,j}(\alpha_a), \qquad D_{Q,j} = \Delta C_{Q,j} - m^{\mathrm{pre}}_{Q,j}(\alpha_b - \alpha_a). \tag{3}$$

The preceding slope is an unweighted linear least-squares fit to the specified preceding incidence samples. Using identical samples and weights for every component and the total preserves additivity for both increments and departures. The primary preceding-incidence range for $\delta = 20°$ is $\alpha \in \{-6, -4, -2\}°$; the early reference $\delta = 30°$ numerical baseline is $\alpha \in \{-16, -14, -12\}°$. These are empirical references, with no assumption that the preceding flow is attached. Neighboring preceding-incidence fitting ranges test this reference dependence, with the same surface solutions. Where the preceding sequence is curved, signed finite differences describe the response without imposing a unique event amplitude.

The uncertainty sensitivity of each measured lift or moment departure is evaluated by propagating the supplied coefficient uncertainties through its defining linear combination. Equation 3 is written as $D = \sum_i w_i C_i$, where i indexes the measured samples, w_i is the combined coefficient weight, and u_i is its supplied propagated uncertainty magnitude. The independence-based sensitivity u_{ind} and absolute-weight sensitivity u_{L1} are

$$u_{\mathrm{ind}} = \left[\sum_i (w_i u_i)^2\right]^{1/2}, \qquad u_{\mathrm{L1}} = \sum_i |w_i| u_i. \tag{4}$$

The first is conditional on independence; the second bounds a perturbation when each sample varies within its supplied magnitude. Coverage and between-incidence covariance are undocumented, so these quantify sensitivity to the supplied uncertainties rather than confidence intervals.

To locate the changes identified by the component load decomposition, pressure and wall-shear forces are integrated

over chordwise and spanwise regions. The pressure coefficient is $C_p = (p - p_\infty)/q_\infty$, and the skin-friction vector is $\mathbf{C}_f = \boldsymbol{\tau}_w/q_\infty$, where p_∞ is freestream pressure and $\boldsymbol{\tau}_w$ is wall shear. Linear surface tractions and their moments are integrated over the triangulated surface, with

$$C_M(x_A) = \frac{1}{q_\infty S\bar{c}} \int \left[z\, dF_x - (x - x_A) dF_z \right]. \tag{5}$$

Here x_A is the moment-reference abscissa. The moment of the local downstream force dF_x is $z\, dF_x$, and that of the local upward force dF_z is $-(x - x_A)dF_z$. These F_x and F_z contributions are retained separately for the common-axis dihedral comparison. Reconstructed full-surface forces and moments agree with the solver-integrated coefficients before decomposition into chordwise and spanwise load contributions; detailed closure checks are retained in the reproducibility records.

The regions distinguish loading upstream of the hinge from loading on the deflected control, and compare inboard, middle, and outboard contributions. The nondimensional span position is $\eta = s/b$, where s is distance along the undeformed span; ξ is the local chord fraction after straightening the control about its hinge. Chordwise regions are forward ($\xi < 0.4$), intermediate ($0.4 \le \xi < 0.7$), and aft ($\xi \ge 0.7$). Chordwise contributions are summed over the complete semispan, while the aft spanwise contributions use semispan thirds (Fig. 3). Contributions include both physical sides, with end-face loads completing the total. Sectional C_p is shown at $\eta = 0.2, 0.5, 0.8$ on both physical sides. The reference sections use the fixed and control panels, excluding narrow hinge-junction and end faces; the dihedral and forward-swept sections use complete surface intersections, including the junction and blunt trailing edge. Curves follow the connected surface intersections without bridging gaps or smoothing. Regional contributions are integrated loads, rather than load densities.

The local downstream direction must account for control rotation and surface orientation before reverse flow is assessed. Let $\mathbf{d}_c$ be the downstream chord direction: $(1, 0, 0)$ on the fixed panel, rotated about the physical hinge on the control. On each surface triangle with unit normal $\mathbf{n}$, its tangent-plane projection is normalized to give $\mathbf{t}_c = [\mathbf{d}_c - (\mathbf{d}_c \cdot \mathbf{n})\mathbf{n}]/\|\mathbf{d}_c - (\mathbf{d}_c \cdot \mathbf{n})\mathbf{n}\|$. This construction points downstream on both physical sides, includes dihedral and flap rotation, and is applied after transforming symmetry-reflected states into the common physical convention. End faces are excluded from the reverse-flow area calculation; the retained upper-control faces have no degenerate projected directions.

The signed coefficient $C_{f,c} = \mathbf{C}_f \cdot \mathbf{t}_c$ is negative where the chordwise component of wall shear points upstream, indicating local reverse flow. We report the fraction f_{rev} of the upper side of the deflected control ($\xi > 0.7$, excluding end faces) satisfying $C_{f,c} < 0$, weighted by actual surface area. Thus $f_{\text{rev}} = 0.5$ means that upstream-directed chordwise shear occupies half that upper-control area. Linear interpolation within each triangle locates the sign crossing. This area fraction measures the extent of reverse flow in the local chordwise direction near the wall. It is not a complete

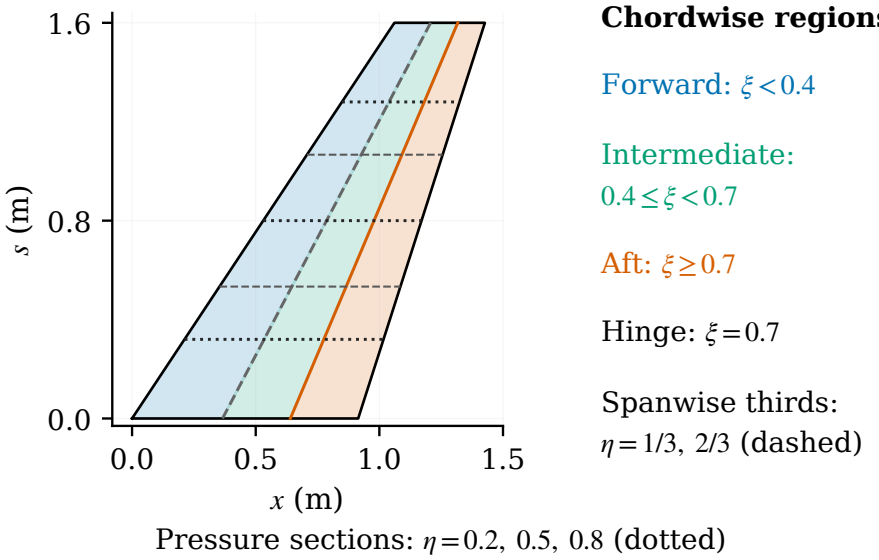


Fig. 3 Undeformed reference planform: chordwise integration regions, spanwise thirds, hinge, and pressure-section locations.

three-dimensional separation footprint: three-dimensional separation is characterized by the topology of the full skin-friction field [15], and on swept wings separation need not be accompanied by flow reversal [16]. The scalar criterion therefore identifies one physically meaningful near-wall signature, without a topological analysis.

III. Experimental–Numerical Comparison

The measurements establish whether lift and pitching moment change together as incidence rises at fixed control deflection. Figure 4 presents the reference zero-dihedral polars. The comparison includes deflections below and above the pronounced reduction in lift growth so that a change in moment response need not be assigned to the same incidence or deflection as a lift feature.

For configuration I at $\delta = 20°$, the lift-coefficient increase over $\alpha = -2 \rightarrow 0°$ is approximately 0.020, compared with 0.089 over the preceding $\alpha = -4 \rightarrow -2°$ interval. This is a roughly 78% reduction in the interval's lift increase, while lift itself continues rising. Moment about the corresponding experimental aerodynamic center increases by +0.036, becoming less negative. Both departures in Table 3, calculated against $\alpha \in \{-6, -4, -2\}°$, exceed the supplied-uncertainty sensitivities. The pointwise lift and moment uncertainty magnitudes are approximately 0.013 and 0.0046–0.0047, respectively; the local incidence spacing is $\Delta\alpha = 2°$.

The two coefficients already respond differently at the lower deflection of $\delta = 10°$: the measured moment change is −0.0038 over $\alpha = -2 \rightarrow 0°$ and +0.011 over $\alpha = 0 \rightarrow 2°$, below the deflection giving the pronounced $\delta = 20°$ reduction in lift growth. At $\delta = 30°$, the early lift feature is broader. The lift-coefficient increases are 0.077, 0.064, and 0.111 over $\alpha = -12 \rightarrow -10°$, $\alpha = -10 \rightarrow -8°$, and $\alpha = -8 \rightarrow -6°$, while moment increases by about 0.018 on each of the first

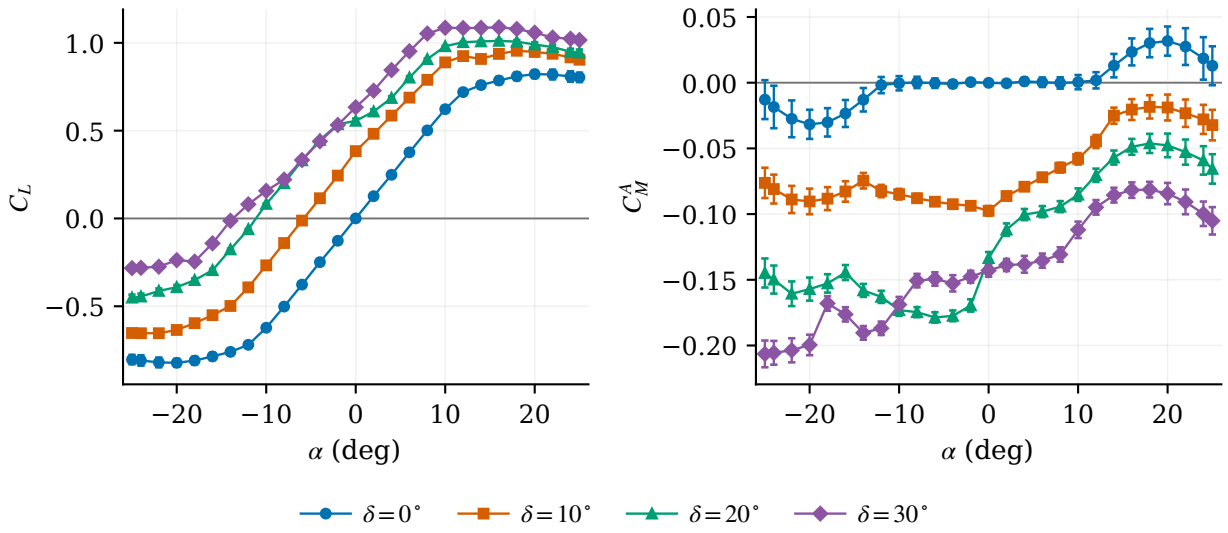


Fig. 4 Experimental lift and pitching moment of the reference planform at zero dihedral for four control-surface deflections.

Table 3 Reference zero-dihedral measured lift and moment departures at $\delta = 20°$, with their uncertainty sensitivities.

Geom., Γ, δ	Coefficient	α interval (deg)	D	u_{ind}	u_{L1}
I, 0°, 20°	C_L	$-2 \rightarrow 0$	-0.08274	0.02357	0.03631
I, 0°, 20°	C_M^A	$-2 \rightarrow 0$	+0.03148	0.00862	0.01360

two intervals. Curvature of the preceding lift sequence makes the departure sensitive to the fitting range, so these actual interval changes describe the early feature more directly.

The numerical comparisons now assess separately the lift and moment evidence available for local interpretation. In Fig. 5, filled/solid and open/dashed curves denote WT and RANS, circles and squares distinguish $\delta = 20°$ and $\delta = 30°$ deflections, and open diamonds mark zero-incidence numerical values. Grey bands mark selected measured incidence intervals, plain for $\delta = 20°$ and hatched for $\delta = 30°$. They are neither uncertainty bands nor exact separation-onset locations. Each dataset's moment is about its corresponding aerodynamic center.

For the reference model at $\delta = 20°$, the computed reduction in lift growth occurs over $\alpha = 0 \rightarrow 2°$, one sampled interval later than in WT. The numerical zero-incidence lift is approximately 0.842, compared with WT 0.556 and uncertainty magnitude 0.013. A qualitatively similar reduction in lift growth and less-negative moment thus coexist with a substantial lift-level bias. At $\delta = 30°$, the early numerical lift change is sharper than the broad measured feature, and the moment departure depends more strongly on the preceding fitting range.

The reference computation captures the principal lift and pitching-moment trends. The comparisons with different

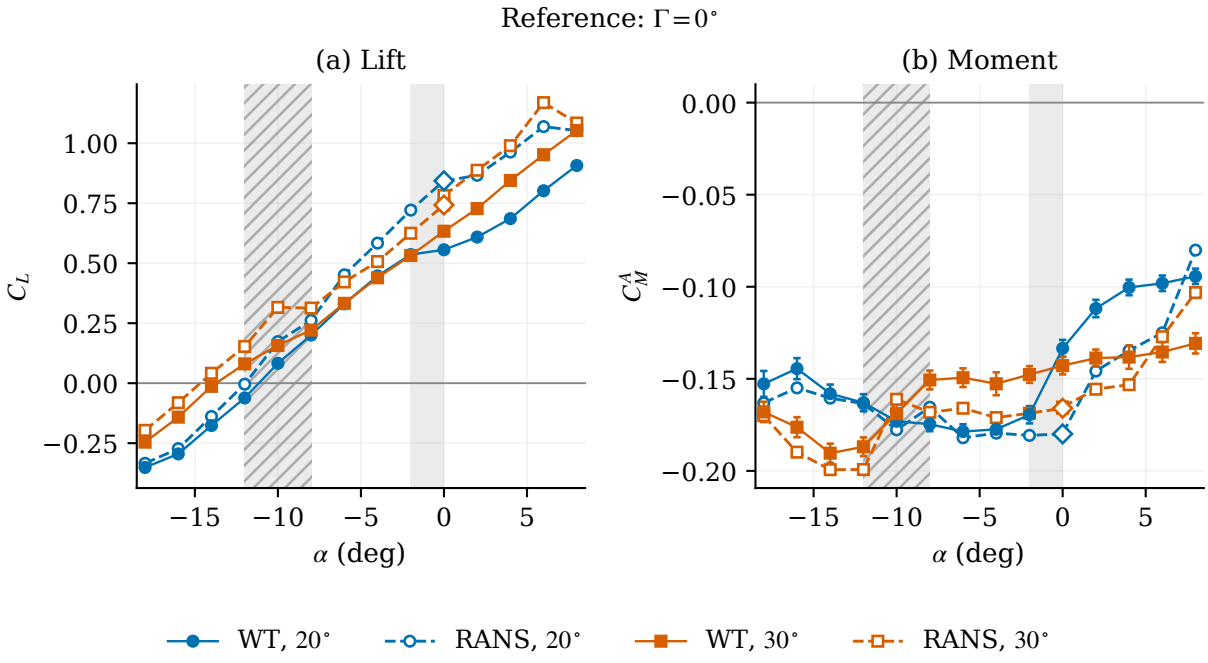


Fig. 5 Reference at zero dihedral: WT and RANS lift and pitching moment. Grey bands: measured intervals, plain $\delta = 20°$, hatched $\delta = 30°$.

geometries are presented with their loading analysis in Sec. V. The following reference analysis investigates the computed loading changes themselves, using the integrated comparison to delimit their experimental interpretation.

IV. Load Breakdown of the Reference Model

A. Fixed-surface and control-surface loads

The reference analysis concerns configuration I at zero dihedral and $\delta = 20°$. In Fig. 6, the upper panels show the fixed, control and total coefficients. Dashed segments continue each preceding fitted slope, anchored at its actual zero-incidence value, across the shaded $\alpha = 0 \rightarrow 2°$ interval. The endpoint gap is the departure; the lower signed markers distinguish the actual increment from that departure. The two displays answer different questions: whether a surface gains load, and how its growth differs from the preceding trend.

The fixed surface actually gains +0.0593 in lift coefficient, while the control loses −0.0351. Their combination leaves a positive total increase of +0.0242. Continued growth of total lift is therefore sustained by the fixed-surface gain exceeding the control loss. Nevertheless, the fixed-surface gain is smaller than expected from its preceding slope, and both surfaces contribute negative departures: −0.0692 fixed and −0.0414 control, with total −0.1106 (Table 4). The reduction in lift growth therefore involves both the fixed surface and the control.

The fixed-surface share of approximately 63% is the ratio $D_{L,\text{fixed}}/D_{L,\text{total}}$, using these two negative departures. It measures each surface's share of the computed lift-growth shortfall. Across the tested preceding-incidence ranges, this

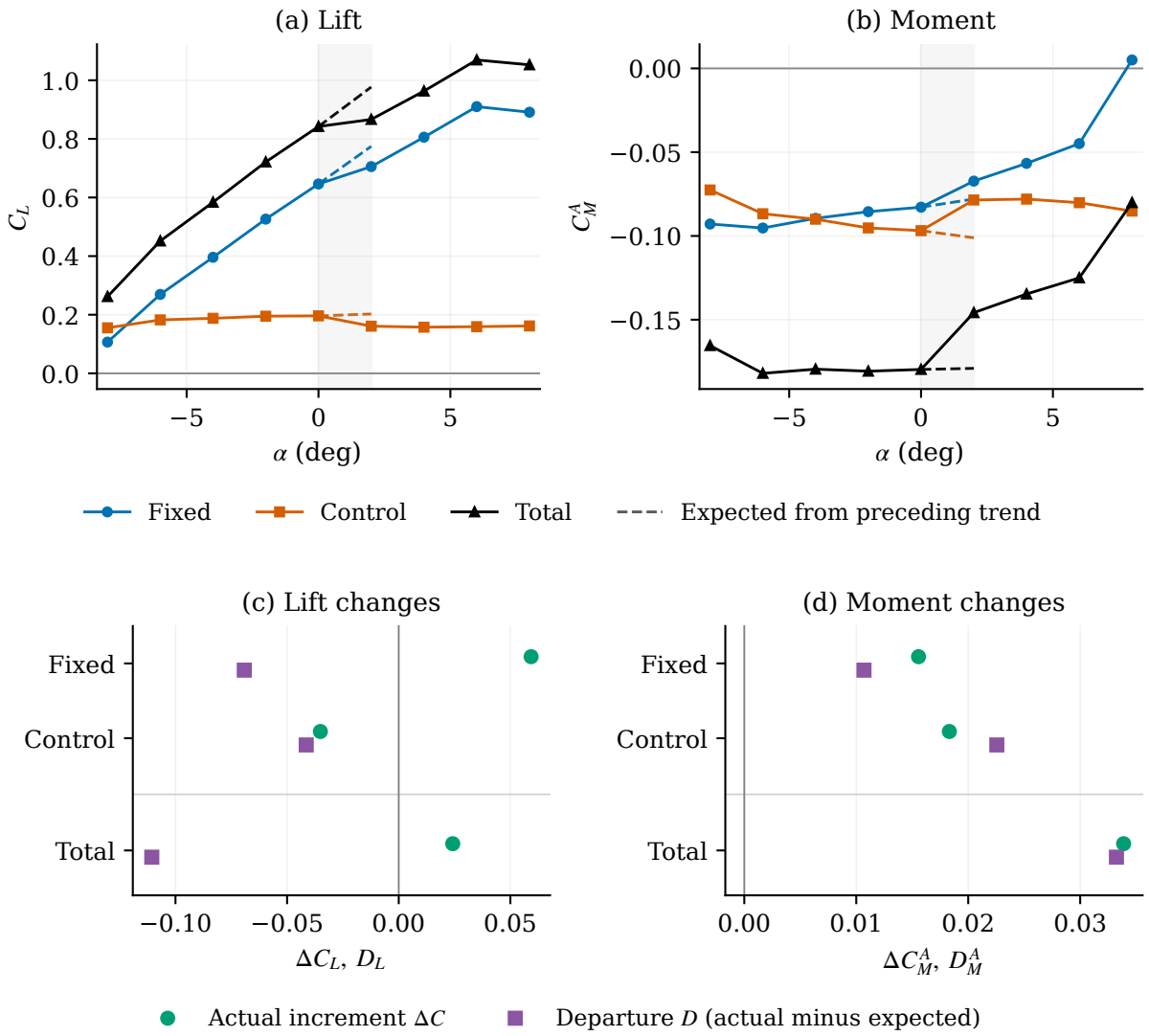


Fig. 6 Fixed-surface, control-surface, and total numerical loading, with actual increments and departures from preceding incidence trends.

fraction remains 62.4–62.6%, while the total lift departure varies from approximately −0.105 to −0.137 (Table 7). The amplitude depends on the empirical preceding trend, but its division between surfaces is comparatively stable.

Pitching moment about the numerical aerodynamic center has different relative contributions from the fixed and control surfaces. The departures reinforce: +0.0107 from the fixed surface and +0.0226 from the control. The fixed surface supplies the larger lift-growth deficit, but the control supplies the larger moment departure. Which contribution is larger depends on the moment axis: at the original numerical origin, fixed/control moment departures are +0.0529/+0.0474, so the fixed contribution is larger. Reinforcement persists at both axes. Locating these changes along the chord and span is therefore necessary to understand the combined force and moment response.

Tip loads contribute less than 0.09% of the changes listed in Table 4 and are included in the totals. The dihedral

Table 4 Reference zero-dihedral lift and aerodynamic-center moment departures from preceding incidence trends; total includes tip loads.

Geom., δ, α interval	Departure	Fixed	Control	Total
I, 20°, $0 \rightarrow 2°$	D_L	-0.0692	-0.0414	-0.1106
I, 20°, $0 \rightarrow 2°$	D_M^A	+0.0107	+0.0226	+0.0332
I, 30°, $-10 \rightarrow -8°$	D_L	-0.0673	-0.0529	-0.1202
I, 30°, $-10 \rightarrow -8°$	D_M^A	-0.0173	+0.0150	-0.0023

load decomposition retains their appreciable contribution explicitly.

B. Pressure distribution and pitching moment

Figure 7 first resolves the local pressure change between the actual $\alpha = 0°$ and $\alpha = 2°$ solutions in the reference load decomposition. Upper and lower sides are overlaid: solid curves with filled markers denote the upper side, and dotted curves with open markers denote the lower side. Circles and squares distinguish incidence. Upper leading-edge suction strengthens, while the suction peak near the hinge weakens. Here, pressure recovery denotes the chordwise rise downstream of the hinge suction minimum. The change in pressure recovery, together with the stronger leading-edge suction, redistributes loading along the chord.

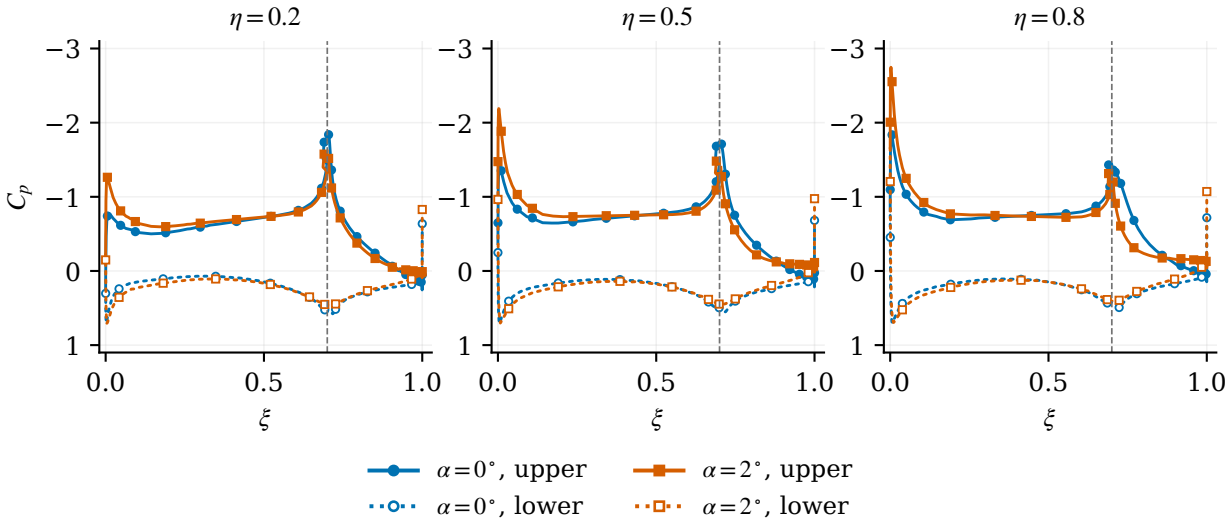


Fig. 7 Chordwise pressure distributions at three span locations across the reference pre-stall loading change.

Figure 8(a) makes the connection to growth by integrating each chord region over the semispan and comparing actual with expected lift increments. The forward region gains +0.0671, consistent with stronger leading-edge suction, but its preceding slope predicts +0.1027; its departure is −0.0357. The intermediate fixed-chord region instead loses −0.0097 rather than gaining the expected +0.0257, giving a departure of −0.0354. Stronger suction near the leading edge

therefore coexists with both reduced forward-region growth and an actual intermediate-region loss. Their combination explains why the complete fixed surface still gains lift while contributing strongly to the reduction in lift growth in Fig. 6.

The aft region loses −0.0332, compared with an expected +0.0063, giving a departure of −0.0396. Pressure supplies nearly all the principal lift departure, while viscous forces remain included in the integration.

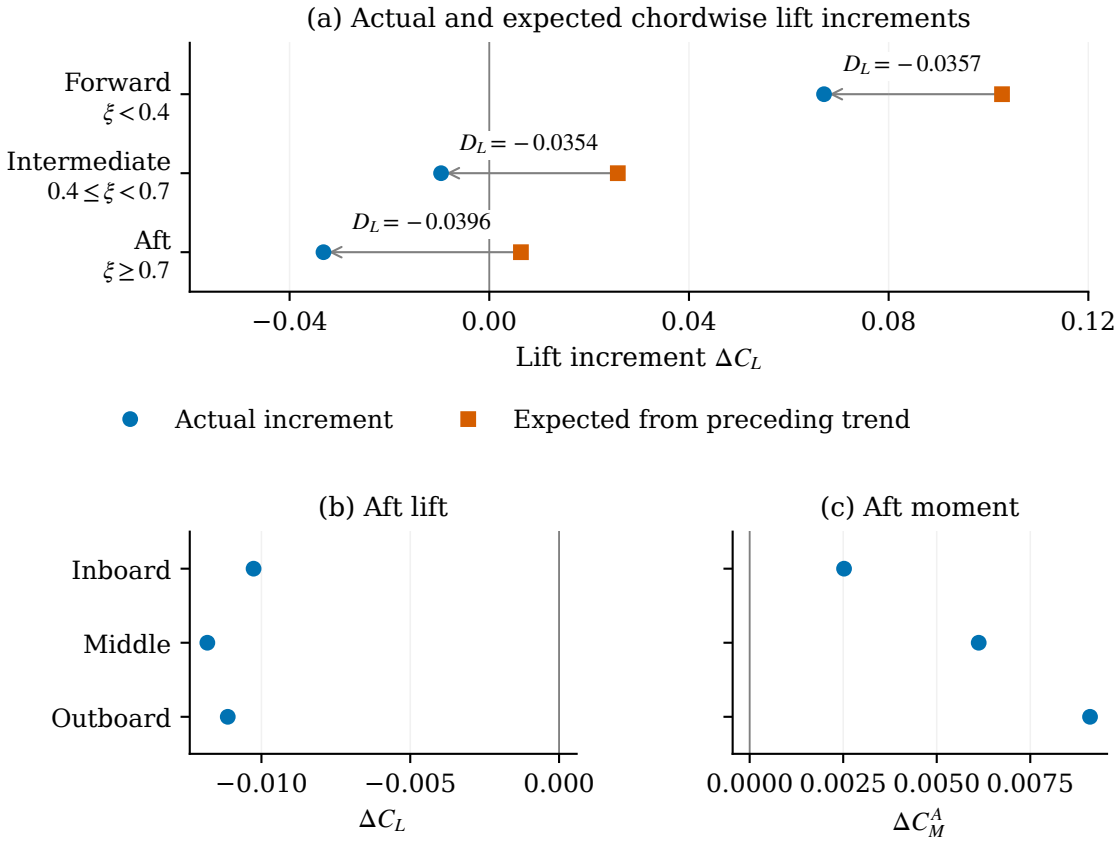


Fig. 8 Chordwise and spanwise contributions to reference lift and pitching-moment changes over the selected pre-stall interval.

Panels (b,c) answer a different question from panel (a): they show actual aft-region ($\xi \ge 0.7$) lift and moment increments in three spanwise regions, with no preceding-trend subtraction. The aft lift changes are similar, −0.0103, −0.0118, and −0.0111 from inboard to outboard, yet moment changes increase from +0.00252 through +0.00612 to +0.00910. The aft-swept geometry in Fig. 3 places outboard aft loading farther downstream of the numerical aerodynamic center. For a local decrease in upward body-normal force, the factor $-(x - x_A)\, dF_z$ gives a positive moment change, larger for a longer downstream arm. The integrated body-normal force decreases in all three aft spanwise regions, supporting this interpretation; their detailed chordwise redistribution also enters the integral. The F_x moment term $z\, dF_x$ remains included and is small here. The longer outboard arms therefore turn comparable aft lift losses into progressively larger positive moment increments.

C. Skin friction and reverse flow

The pressure and load changes can now be compared with the evolution of near-wall flow on the deflected control. Figure 9 reports the proportion of its upper surface on which chordwise wall shear points upstream, using the definition in Sec. II.D. Only the primary $C_{f,c} < 0$ measure is plotted. The open diamond marks $\alpha = 0°$, the start of the interval examined in the surface maps and load analysis.

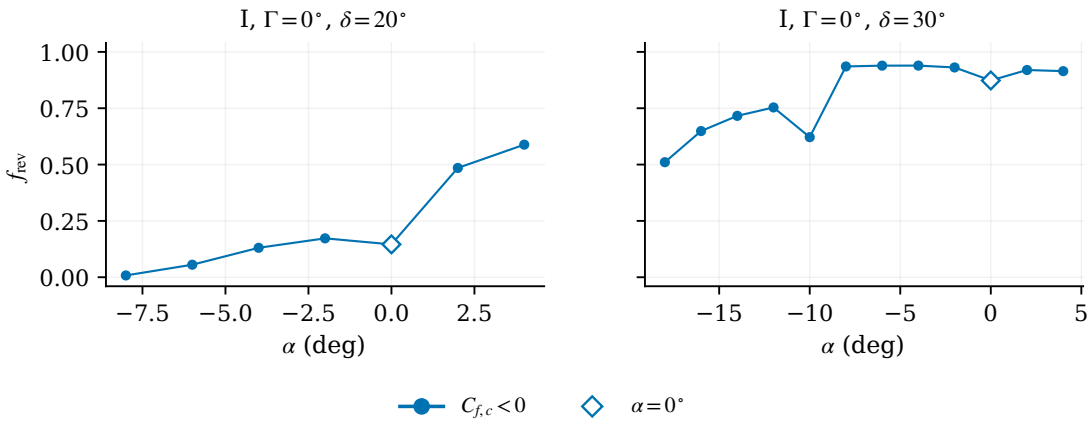


Fig. 9 Reference zero-dihedral upper control surface area fraction with reverse flow at $\delta = 20°$ and $\delta = 30°$.

For the reference $\delta = 20°$ calculation, that area fraction grows from about 15% to 49% (0.146 to 0.485) across $\alpha = 0 \rightarrow 2°$, the same interval as the reduced fixed-surface growth, control lift loss and positive moment change. The maps in Fig. 10 distinguish this increase in area from a change in location.

Figure 10 shows the signed coefficient $C_{f,c}$ on the upper surface for both reference deflections, using identical projections and color scales. At $\delta = 20°$, the $\alpha = 0°$ state has a narrow region of reverse flow adjacent to the hinge, followed by predominantly downstream shear. At $\alpha = 2°$, the main region of reverse flow instead lies near the trailing edge. At $\alpha = 4°$, this region of reverse flow near the trailing edge extends farther upstream. The change across $\alpha = 0 \rightarrow 2°$ therefore includes a change in the location of reverse flow, alongside the weakened hinge suction and aft load loss in Figs. 7 and 8; it is not simply uniform growth of the initial hinge-adjacent band.

By $\alpha = 4°$, about 59% of the upper control experiences reverse flow, while the total lift increase over $\alpha = 2 \rightarrow 4°$ is 0.0968, larger than the preceding 0.0242. Stronger total lift growth can thus coexist with a larger area of reverse flow because the integrated response combines the evolving loading on both surfaces. The extent of reverse flow characterizes near-wall flow, but does not by itself determine the pressure force or pitching moment.

At $\delta = 30°$, the reference reduction in lift growth occurs much earlier, over $\alpha = -10 \rightarrow -8°$. Fixed lift still increases by +0.0406, while the control loses −0.0436, leaving a small total decrease of −0.0030. Both lift departures remain negative. The lower row of Fig. 10 follows this earlier interval. Over it, the upper control surface reverse-flow area fraction rises from approximately 62% to 94% (0.622 to 0.935). The recurring association is thus reduced aft loading

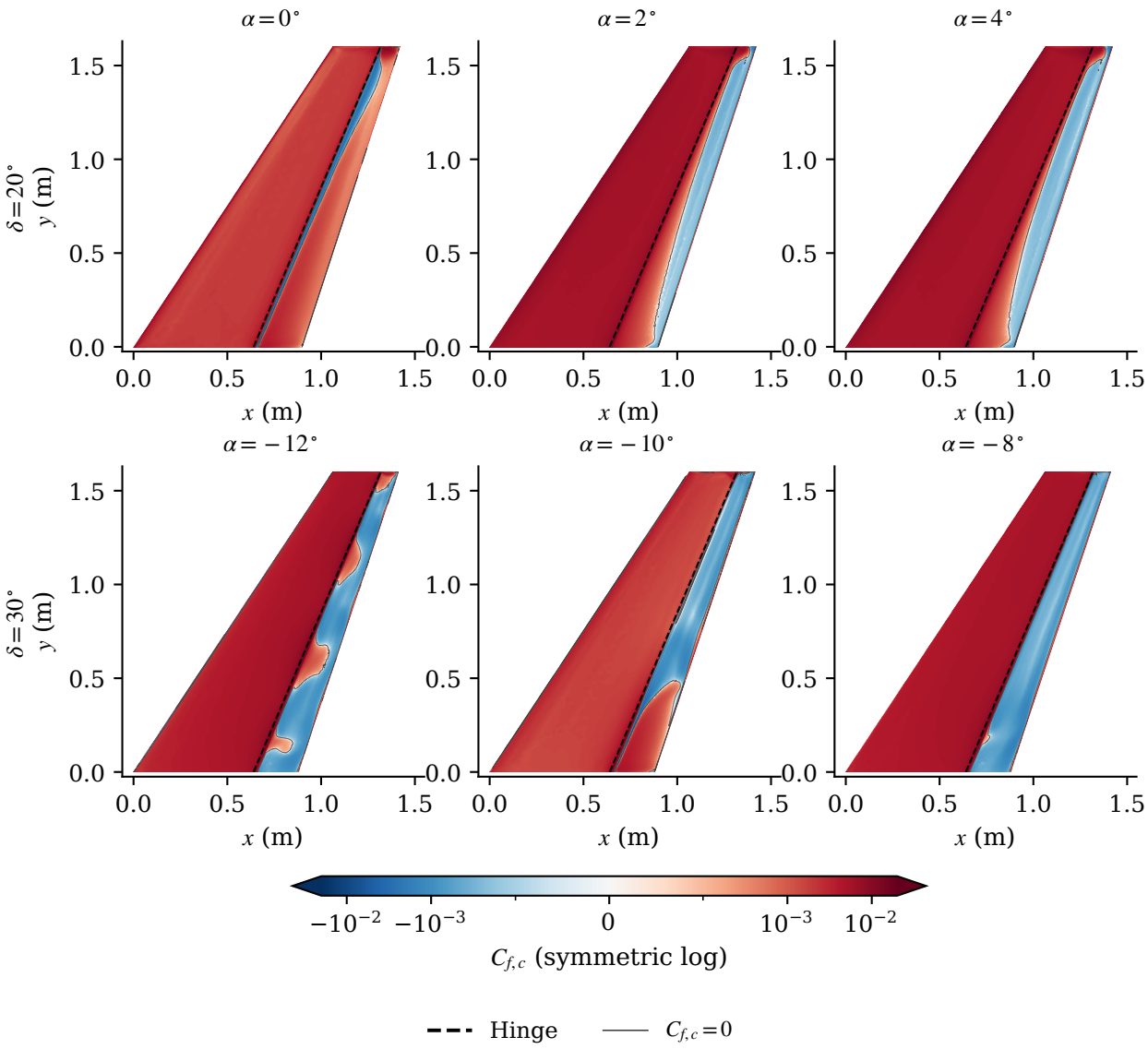


Fig. 10 Reference upper-surface chordwise skin friction across the loading changes at $\delta = 20°$ (top) and $\delta = 30°$ (bottom) deflection.

with expansion of the existing reverse-flow region, although the total lift increment changes sign because the fixed gain no longer outweighs the control loss.

The moment result at $\delta = 30°$ is different: the fixed-surface departure is negative and the control departure positive, so they partly cancel (Table 4). Across the tested preceding fitting ranges, total moment departure varies from approximately -0.0262 to $+0.0076$ as the opposing contributions change their balance. Its total sign is therefore not robust to this empirical reference. This contrasts with the reinforcing moment departures at $\delta = 20°$ and shows why reverse-flow area alone is insufficient to infer the sign of the integrated pitching response.

V. Effects of Dihedral and Sweep

The same component and regional load decomposition is now applied to the dihedral and planform comparisons. Their measured responses and WT–RANS correspondence are presented beside the numerical loading analysis.

Table 5 reports the measured departures and uncertainty sensitivities for these geometry comparisons, using the definitions in Sec. II.D.

Table 5 Measured lift and moment departures for the dihedral and forward-swept cases, with their uncertainty sensitivities.

Geom., Γ, δ	Coefficient	α interval (deg)	D	u_{ind}	u_{L1}
I, 45°, 20°	C_L	$-2 \rightarrow 0$	-0.04785	0.01694	0.02634
I, 45°, 20°	C_M^A	$-2 \rightarrow 0$	+0.00847	0.02630	0.04082
I, 45°, 30°	C_L	$-2 \rightarrow 0$	-0.04038	0.01646	0.02996
I, 45°, 30°	C_M^A	$-4 \rightarrow -2$	-0.02860	0.02478	0.03822
II, 0°, 20°	C_L	$0 \rightarrow 2$	-0.06902	0.02090	0.03739
II, 0°, 20°	C_M^A	$0 \rightarrow 2$	+0.01668	0.00793	0.01424

A. Effect of dihedral

Figure 11 brings together the wind-tunnel dihedral comparison (upper panels) and the $\Gamma = 45°$ WT–RANS comparison (lower panels). At $\delta = 20°$, measured lift growth remains suppressed across $\alpha = -2 \rightarrow 2°$, while the moment departure over $\alpha = -2 \rightarrow 0°$ is smaller than both uncertainty sensitivities in Table 5. At $\delta = 30°$, the selected reduction in lift growth occurs over $\alpha = -2 \rightarrow 0°$, but moment decreases earlier, by 0.027 over $\alpha = -4 \rightarrow -2°$. That moment departure is comparable with u_{ind} and smaller than u_{L1}. The joint response differs from the clearly resolved reference $\delta = 20°$ lift/moment change.

The computed sequence has weaker correspondence with these measurements. At $\delta = 20°$, numerical lift decreases over $\alpha = -6 \rightarrow -4°$ while measured lift increases; RANS does not convincingly reproduce the sustained measured reduction in lift growth or its weak moment feature. At $\delta = 30°$, the negative numerical moment change occurs over $\alpha = -2 \rightarrow 0°$, one sampled interval later than the measured change. The following decomposition consequently concerns the computed dihedral response.

At $\alpha = 0°$, $\delta = 20°$, the computed lift coefficient is approximately 0.301 lower with $\Gamma = 45°$. Projection of the reference panel-normal loading onto the common lift direction accounts algebraically for about 0.247 of this difference; changes in local loading and the spanwise-force contribution supply the remainder. The absolute lift difference therefore includes a substantial projection contribution, while the successive incidence increments require the separate load analysis below.

The computed lift increments change differently with incidence on the two models. At the same $\delta = 20°$ deflection, the actual lift increase over $\alpha = -2 \rightarrow 0°$ is 0.1210 at zero dihedral and 0.0763 at $\Gamma = 45°$. Across the next interval,

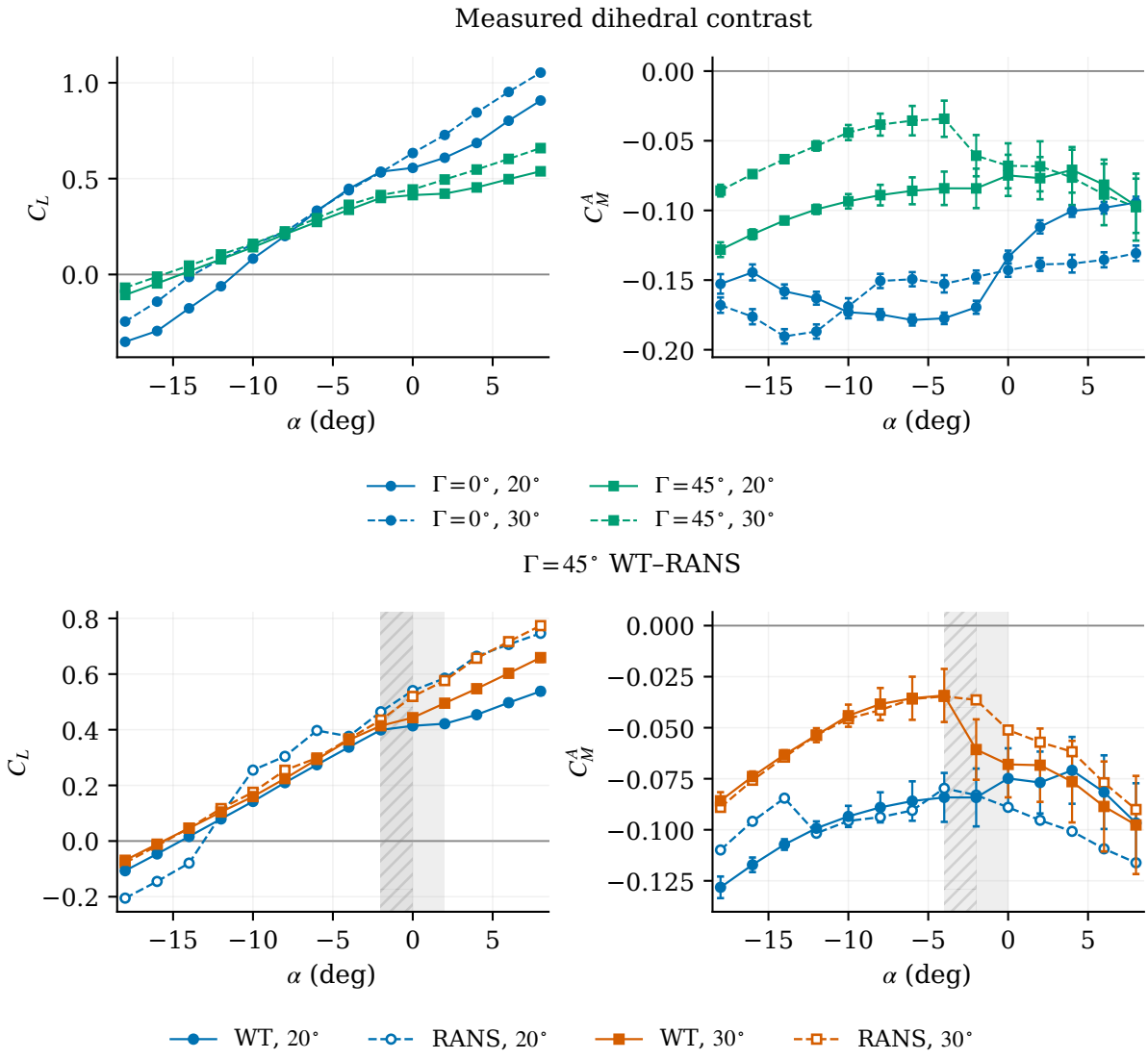


Fig. 11 Wind-tunnel dihedral comparison (top) and positive-dihedral WT–RANS comparison (bottom). Grey bands mark measured intervals: plain $\delta = 20°$, hatched $\delta = 30°$.

$\alpha = 0 \rightarrow 2°$, the ordering reverses: the increases are 0.0242 and 0.0444. Although the dihedral model has lower absolute lift, it gains more lift over that latter interval. The change in ordering shows that the relative lift growth depends on incidence. Table 6 shows that the $\Gamma = 45°$ fixed surface still gains lift over $\alpha = 0 \rightarrow 2°$, while the control loses lift. Its moment about its own computed aerodynamic center decreases through contributions from the fixed surface, control and tip. The table retains the tip explicitly because its moment contribution is appreciable in these dihedral cases.

Pitching moment depends on where F_x and F_z act. Dihedral places the loading above the reference axis, so F_x contributes through the vertical arm z. The F_z contribution is weighted by the downstream arm $x - x_A$ (Eq. 5). Both numerical solutions are referred to the same physical axis A_I. Maggin and Shanks [17] showed why this choice matters:

Table 6 Actual coefficient increments for the dihedral and forward-swept cases; moments are about their corresponding numerical aerodynamic centers.

Geom., Γ, δ, α interval	Increment	Fixed	Control	Tip	Total
I, 45°, 20°, 0 → 2°	ΔC_L	+0.0532	-0.0106	+0.0018	+0.0444
I, 45°, 20°, 0 → 2°	ΔC_M^A	-0.0044	-0.0003	-0.0017	-0.0063
I, 45°, 30°, −2 → 0°	ΔC_L	+0.0864	-0.0028	+0.0013	+0.0848
I, 45°, 30°, −2 → 0°	ΔC_M^A	-0.0112	-0.0025	-0.0011	-0.0148
II, 0°, 20°, 0 → 2°	ΔC_L	+0.1014	-0.0079	0.0000	+0.0935
II, 0°, 20°, 0 → 2°	ΔC_M^A	+0.0112	+0.0057	0.0000	+0.0168

for a 40° swept-back wing, moments about fixed reference axes indicated an apparent increase in longitudinal stability with geometric dihedral, whereas only a slight change remained when each configuration was referred to its own mean aerodynamic chord. The common axis therefore compares the loading and its moment arms about one physical point, whereas Table 6 uses each configuration's computed aerodynamic center. Figure 12(a,b) compares their moment increments through

$$\mathcal{I}_M = [C^{A_I}_{M,45}(\alpha_b) - C^{A_I}_{M,45}(\alpha_a)] - [C^{A_I}_{M,0}(\alpha_b) - C^{A_I}_{M,0}(\alpha_a)] \tag{6}$$

A negative $\mathcal{I}_M$ means that the moment increase is smaller, or the decrease larger, with dihedral over the same incidence interval. These are differences of actual moment increments, not departures from preceding fitted trends.

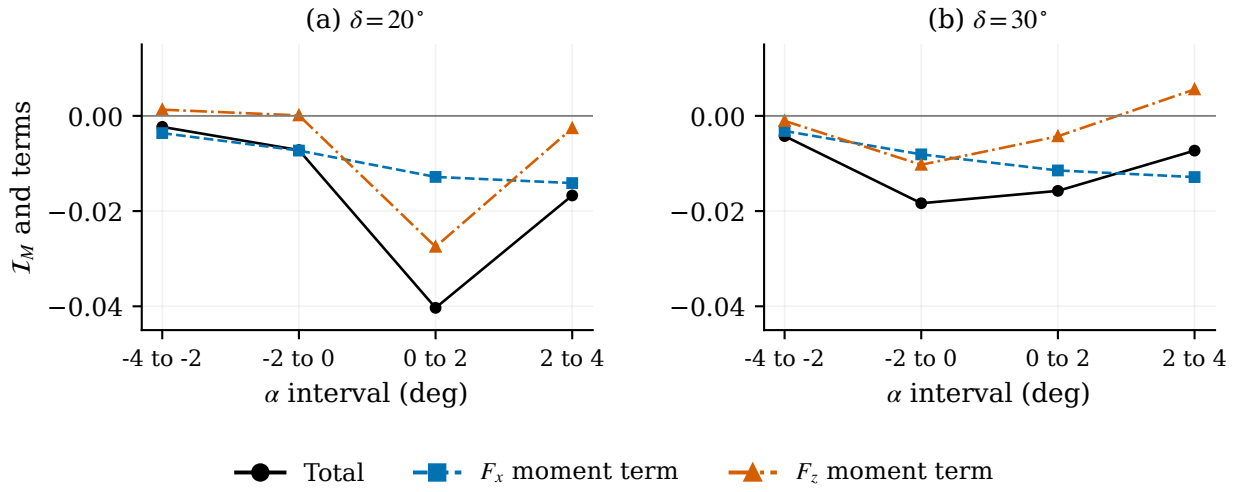


Fig. 12 Dihedral differences in moment increments about the common axis A_I and their F_x and F_z contributions: $\delta = 20°$ (left) and $\delta = 30°$ (right).

At $\delta = 20°$, the F_x moment term gives a negative difference in every plotted interval. Over $\alpha = -4 \rightarrow -2°$, it contributes −0.00362, partly offset by a +0.00131 contribution from F_z, leaving $\mathcal{I}_M = -0.00231$. Across $\alpha = 0 \rightarrow 2°$, the terms instead reinforce: −0.01283 from F_x and −0.02750 from F_z give −0.04033. The F_z moment term supplies

the larger difference during this reference loading change. Over $\alpha = 2 \rightarrow 4°$, its contribution falls to -0.00260, while the contribution from F_x remains -0.01412 and dominates the -0.01672 total. A single relative weighting of the two moment terms therefore cannot describe the sequence.

At $\delta = 30°$, the terms reinforce over $\alpha = -2 \rightarrow 0°$: -0.00808 from F_x and -0.01027 from F_z give a total -0.01836. Over $\alpha = 0 \rightarrow 2°$, both remain negative, with the contribution from F_x larger. Over $\alpha = 2 \rightarrow 4°$, a positive difference from F_z of $+0.00556$ offsets part of the -0.01286 difference from F_x, leaving -0.00730. This change from reinforcement to compensation concerns the moment-term differences between geometries; it is distinct from fixed/control moment-departure reinforcement in Sec. IV.A.

The moment changes raise a further question: do they accompany expansion of reverse flow on the control, or loading changes elsewhere on the model? At $\delta = 30°$, the upper-control reverse-flow area fraction is already 0.889 at $\alpha = -2°$, increasing only to 0.911 and 0.912 at $\alpha = 0°$ and $\alpha = 2°$ (Fig. 13). Over $\alpha = -2 \rightarrow 0°$, the moment about this configuration's computed aerodynamic center nevertheless decreases by 0.0148, with the fixed surface supplying -0.0112 of that increment (Table 6). This combination motivates examining the fixed-surface pressure and skin friction over the same incidence sequence.

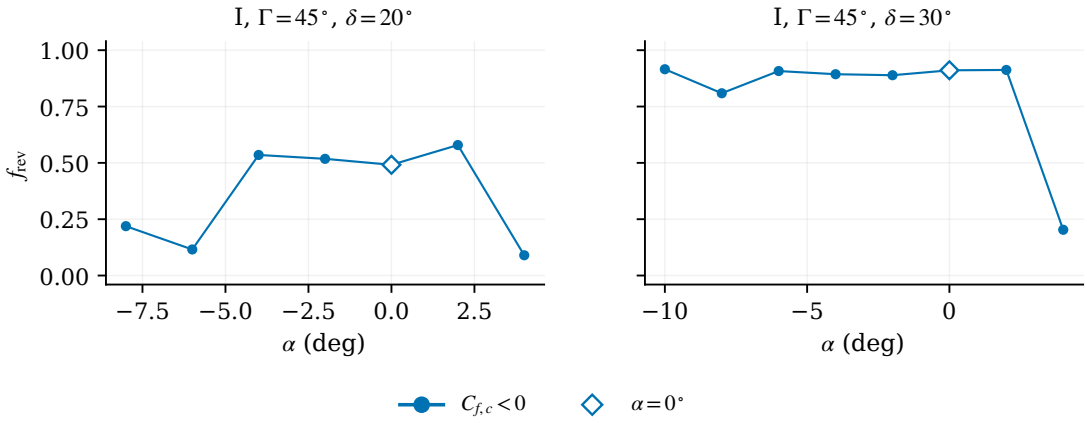


Fig. 13 Reverse-flow area fractions on the upper control surface for $\Gamma = 45°$ at $\delta = 20°$ and $\delta = 30°$.

Figure 14 relates this extensive reverse flow over the control surface to the changing flow over the fixed surface. The middle row adds upper-surface $C_{f,c}$ cuts at the same span stations as the pressure sections. Between $\alpha = -2°$ and $\alpha = 0°$, a band of increased chordwise shear develops from approximately midspan toward the tip. This abrupt rise in $C_{f,c}$ is transition-like. At $\alpha = 2°$, a narrow leading-edge region of upstream-directed chordwise shear appears over roughly the outer third of the span. At $\eta = 0.8$, this short region of negative shear is followed by recovered downstream shear and a sharp increase in its magnitude. Together with the local pressure distribution, the pattern is compatible with a local separation/reattachment process and could indicate a transition-related change in the computed boundary layer. The surface data do not independently identify the laminar–turbulent transition location or a complete three-dimensional

bubble topology. On an unswept finite wing at a much lower Reynolds number, Toppings and Yarusevych [18] measured substantial spanwise variation in separation-bubble length, thickness, and reverse-flow velocity, with an open topology and eventual suppression near the tip. Spanwise variation of such features is therefore plausible on a finite wing, although those measurements do not define the present swept-wing state.

The pressure sections show the accompanying spanwise variation of loading. At $\alpha = 0°$, the leading-edge suction is weak at $\eta = 0.2$, whereas distinct peaks appear at $\eta = 0.5$ and 0.8. The hinge-region suction also changes with span, with stronger peaks at the middle and outboard cuts. Stronger suction followed by pressure recovery gives the outboard boundary layer a different pressure-gradient history, consistent with the shear rise and leading-edge reverse flow appearing there first. Spanwise variation of local inflow provides a possible contribution to the suction difference. Taper reduces the local chord and, for the same section shape, the dimensional leading-edge radius toward the tip. The leading-edge acceleration can then occur over a shorter physical distance, increasing the spatial acceleration and pressure gradient under otherwise comparable conditions. This tendency is consistent with the stronger outboard suction, while the computed pressure distribution also reflects the coupled three-dimensional flow and the spanwise change in chord Reynolds number. Both dihedral configurations share this taper, so it can contextualize the spanwise variation within each model but does not by itself explain the difference produced by dihedral.

Over most of the control chord, the pressure curves change little between $\alpha = -2°$ and $\alpha = 2°$, consistent with the nearly unchanged area of reverse flow. The fixed-surface shear and pressure changes therefore occur while the control remains extensively separated. The component increments in Table 6 and the F_x/F_z terms in Fig. 12(a,b) show how this loading produces the computed moment decrease.

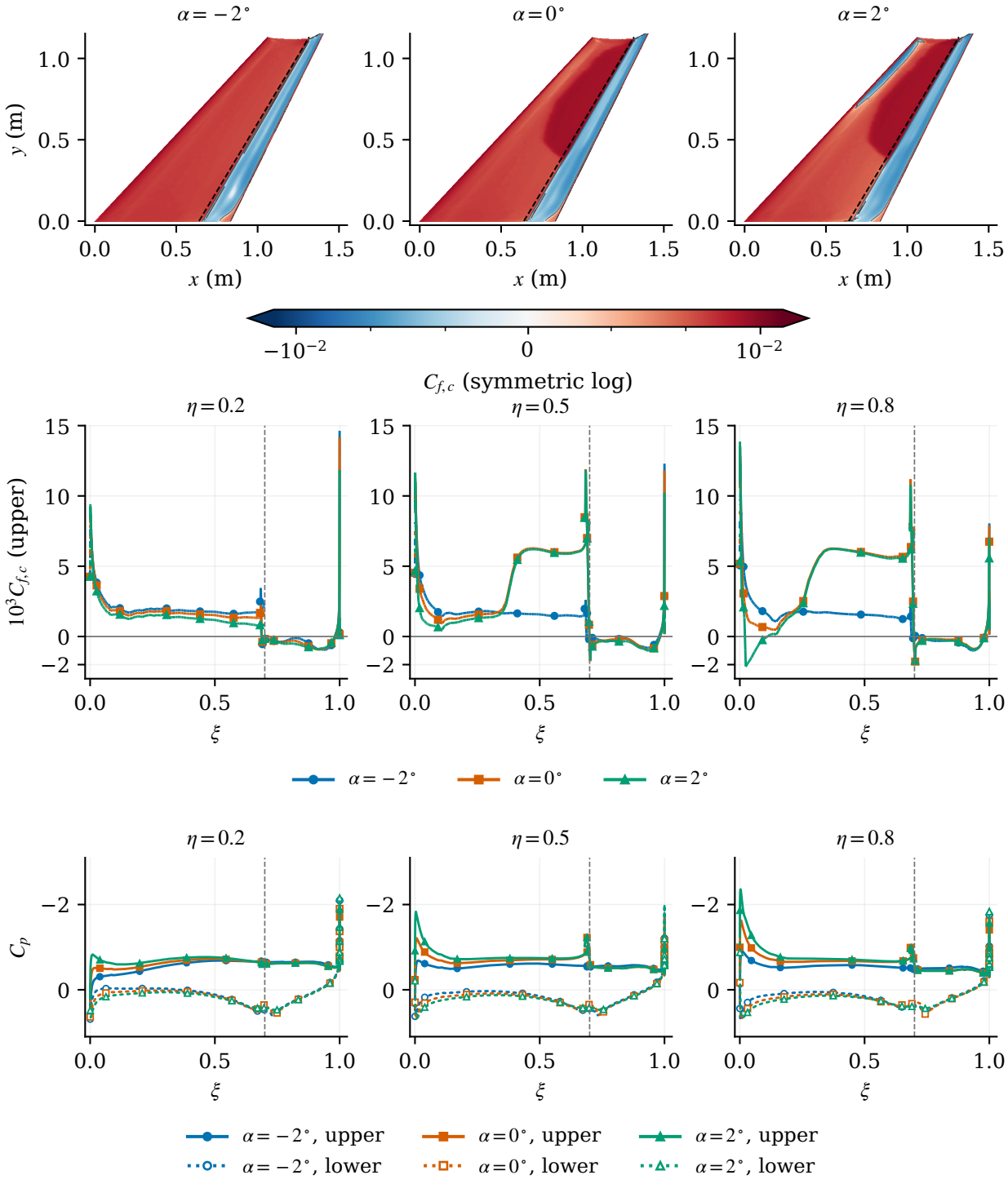


Fig. 14 $\Gamma = 45°$, $\delta = 30°$**: upper-surface shear maps (top), upper-surface shear cuts (middle), and pressure sections (bottom) at** $\alpha = -2, 0, 2°$**.**

B. Effect of sweep direction

The second comparison reverses quarter-chord sweep from $\lambda = +30^\circ$ to $\lambda = -15^\circ$ while retaining the 1.6 m semispan and full-span 30%-chord control. Sweep is the principal geometric contrast. Area and MAC differ by approximately 4% and 8%, respectively, and taper changes from 0.4 to 0.6 (Table 1); these accompanying differences prevent attributing the response to sweep alone.

The wind-tunnel results in Fig. 15 show configuration II's local reduction in lift growth at $\delta = 20^\circ$ over $\alpha = 0 \rightarrow 2^\circ$: lift increases by 0.044, compared with approximately 0.092 over both $\alpha = -2 \rightarrow 0^\circ$ and $\alpha = 2 \rightarrow 4^\circ$. Moment increases by 0.019 over $\alpha = 0 \rightarrow 2^\circ$. This feature is later than on configuration I, with a smaller moment rise across its own interval. The primary moment departure in Table 5 exceeds both uncertainty sensitivities, although neighboring preceding fitting ranges do not uniformly exceed the primary L1 sensitivity. At $\delta = 30^\circ$, the reference model's broad early feature is not repeated at the same negative incidences by II.

RANS gives an $\alpha = 0 \rightarrow 2^\circ$ lift increase of 0.0935 at $\delta = 20^\circ$, more than twice the measured value. Its moment increase, 0.0168, is locally close to the measured 0.019, with the different aerodynamic-center positions retained. This weaker lift correspondence limits the use of its local solution to explain the measured feature.

Figure 16 shows how the two planforms redistribute loading over the same $\alpha = 0 \rightarrow 2^\circ$ interval at $\Gamma = 0^\circ$, $\delta = 20^\circ$. The forward-swept fixed surface gains +0.1014 in lift coefficient, compared with +0.0593 on I, while its control loses only −0.0079 rather than −0.0351. The resulting total lift increase is larger on II, +0.0935 versus +0.0242, but its pitching-moment increase is smaller, +0.0168 versus +0.0339. Each planform retains its own reference area and computed aerodynamic center. The chordwise loads locate the lift difference: the forward region gains more lift on II, the intermediate region gains +0.0063 instead of losing −0.0097, and the aft loss is reduced from −0.0332 on I to −0.0065 on II. The smaller intermediate and aft moment increments on II, with similar forward-region contributions, accompany this redistribution.

The pitching response depends additionally on where the aft force changes occur relative to the moment axis. On I, comparable aft lift losses occur across the span, yet the positive moment increments increase toward the tip. On II, aft lift loss and positive moment change are concentrated inboard, while the outboard aft region gains lift. The aft regions lie downstream of the respective numerical aerodynamic centers, but the streamwise arm $x - x_A$ increases outboard on I and decreases on II. Consequently, the $-(x - x_A)dF_z$ term in Eq. 5 gives an equal loss of upward force a larger positive moment contribution outboard on I. These different spanwise force changes and moment arms explain why the larger total lift increase on II does not produce a larger pitching-moment increase. McCormack and Cook [16] documented a related role of inboard chordwise load location on a 45° forward-swept wing without controls. In its moderate-lift range, failed aft pressure recovery over the inboard sections shifted their centers of pressure rearward with little change in section lift, spreading outboard with incidence and changing the wing pitching moment before the lift-curve slope decreased. That stall study differs in section, Reynolds number, and control configuration, but similarly shows that the

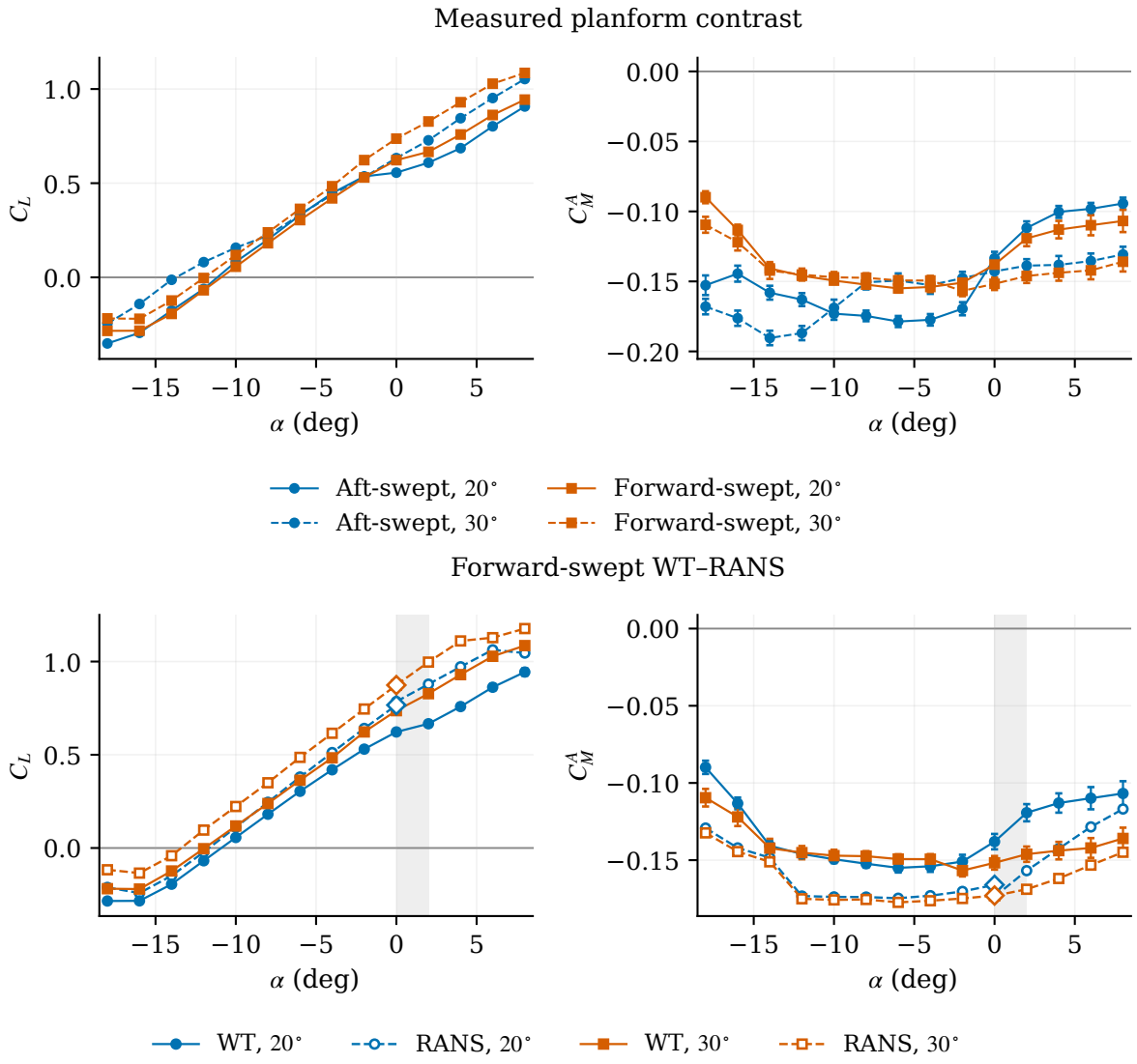


Fig. 15 Wind-tunnel planform comparison (top) and forward-swept WT–RANS comparison (bottom). Grey band marks the measured $\delta = 20°$ interval.

chordwise and spanwise location of inboard load changes can govern the forward-swept pitching response.

Actual lift growth can still fall short of its preceding trend. The fixed/control lift departures are −0.0263/−0.0104 on II, compared with −0.0692/−0.0414 on I; even II's positive intermediate-region increment represents reduced growth relative to that trend. Moment departures reinforce about the computed aerodynamic centers, with the control contribution larger on I and the two contributions similar on II. At II's original numerical origin, however, the fixed/control departures are −0.0023/+0.0035 and partly cancel. The following shear and pressure distributions locate the control-surface flow changes accompanying these integrated differences.

Figure 17 places the increase in II's reverse-flow area fraction beside its loading analysis. The upper-control fraction

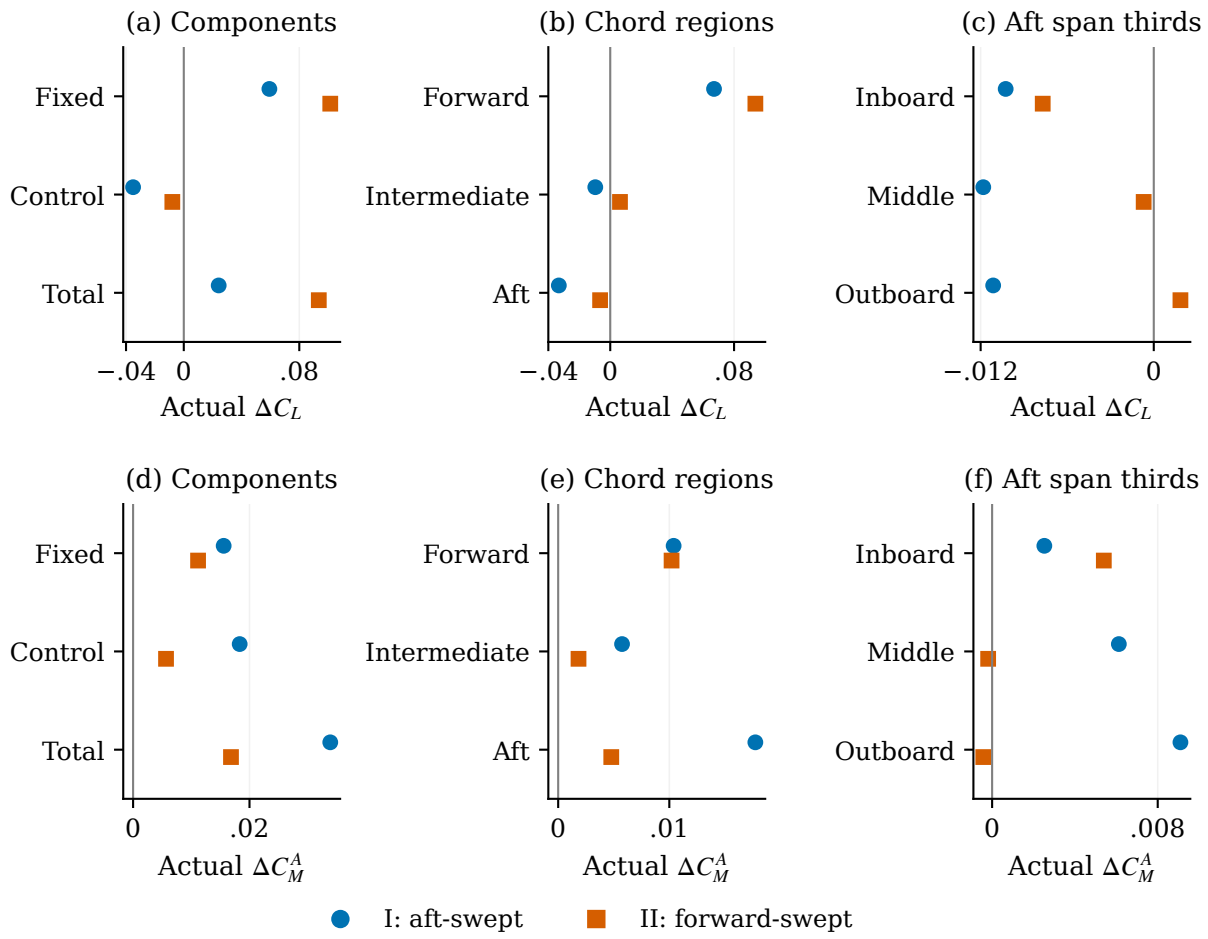


Fig. 16 Planform comparison over $\alpha = 0 \rightarrow 2°$, $\delta = 20°$: lift increments (top) and aerodynamic-center moment increments (bottom), with matching decompositions.

grows from about 19% to 47% and 59% at $\alpha = 0, 2, 4°$. The surface maps in Fig. 18 locate this change: at $\alpha = 0°$, reverse flow near the trailing edge is concentrated around midspan; with increasing incidence it extends upstream along the control chord and toward the root. The pressure sections across $\alpha = 0 \rightarrow 2°$ show a weakened upper-side hinge-suction peak inboard and comparatively little change at the outboard cut, consistent with the concentration of aft lift loss inboard.

At $\alpha = 2°$, II's reverse-flow area fraction is close to I's 49%, despite its much smaller control lift loss across $\alpha = 0 \rightarrow 2°$. The differing pressure distributions and load locations explain why comparable areas of reverse flow can accompany different lift and moment changes. Together, the two planform comparisons show how fixed-surface growth, control loss and their chordwise/spanwise locations combine in the integrated response.

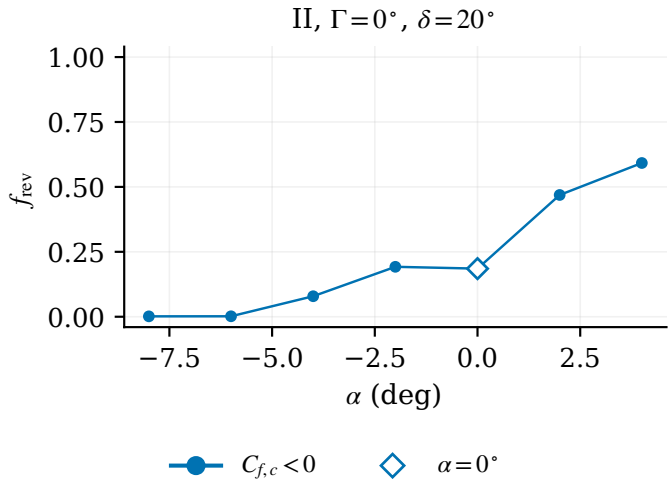


Fig. 17 Forward-swept upper control surface reverse-flow area fraction at $\delta = 20°$; the open diamond marks $\alpha = 0°$, the start of the interval in the loading analysis.

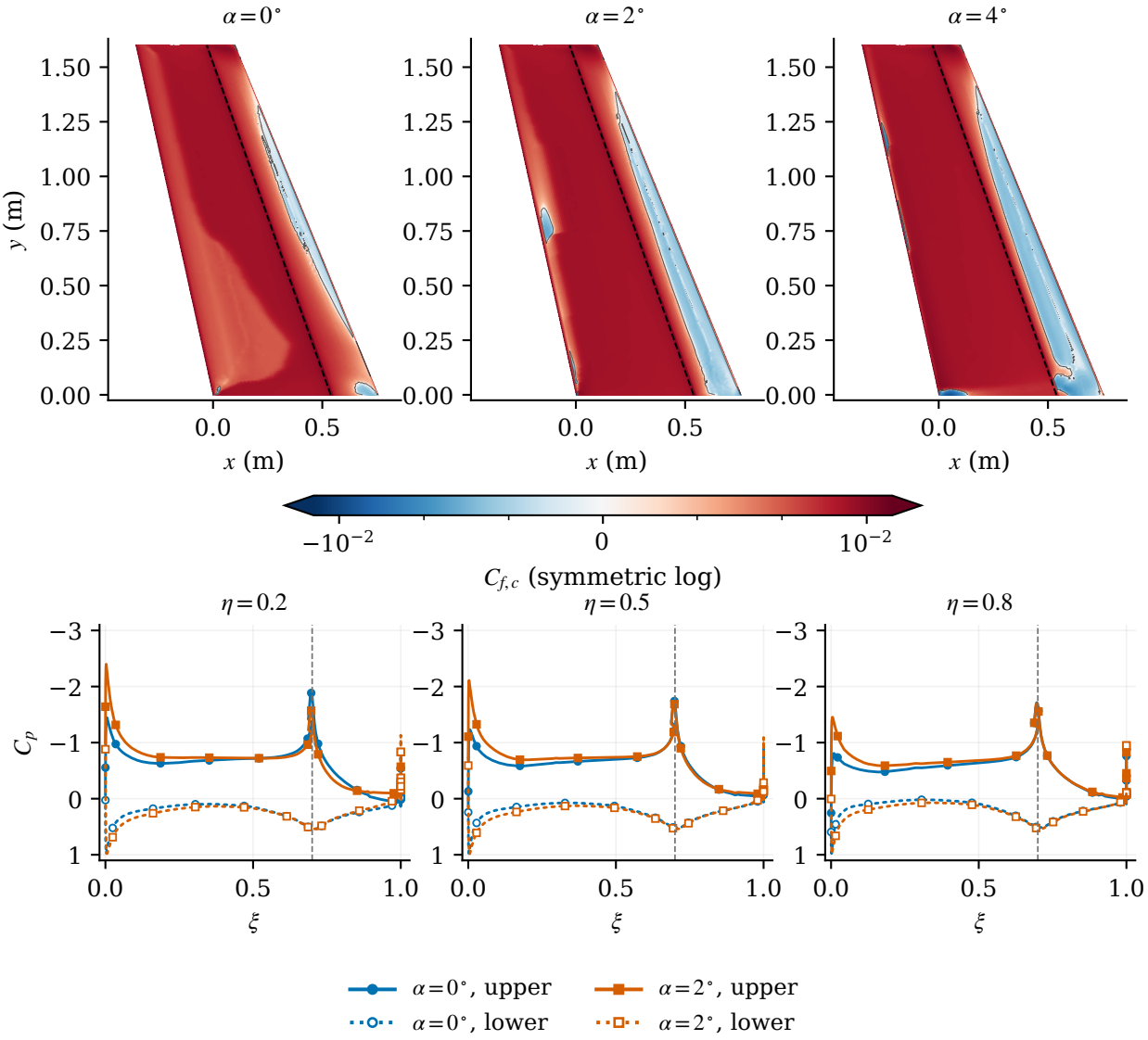


Fig. 18 Forward-swept planform, $\delta = 20°$: upper-surface chordwise shear (top) and pressure sections across $\alpha = 0 \rightarrow 2°$ (bottom).

VI. Conclusions

The wind-tunnel measurements show that reductions in pre-stall lift growth and changes in pitching moment do not necessarily occur over the same incidence interval or at the same control deflection. Resolving the computed forces into fixed- and control-surface contributions, then examining their spatial distribution, identifies how different combinations of loading changes produce the numerical lift and pitching responses.

The reference computation shows that continued total lift growth can conceal reduced growth on both the fixed surface and the full-span control. For configuration I at zero dihedral and $\delta = 20°$, fixed lift gain exceeds actual control lift loss, yet the fixed surface accounts for approximately 63% of the total departure below the preceding trend. The control contributes more strongly to the positive pitching-moment departure about the numerical aerodynamic center.

Pressure sections and regional integrations explain why stronger leading-edge suction is consistent with reduced fixed-surface growth: the forward-region gain falls below its preceding trend, while the intermediate region actually loses lift. Similar aft lift losses produce larger positive moment increments outboard as their downstream moment arms grow. The upper-control shear pattern changes from a region of reverse flow adjacent to the hinge to reverse flow near the trailing edge across the computed interval. Its subsequent upstream extension coexists with renewed total lift growth. The extent of reverse flow consequently complements the pressure and load evidence without determining the integrated response by itself.

The measurements establish distinct lift and moment responses across deflection and geometry. RANS reproduces the principal reference trends with a displaced lift interval and substantial coefficient-level bias, while correspondence is weaker for the dihedral and forward-swept cases. The detailed loading findings therefore characterize the computed flow fields and loads.

For matched dihedral, adjacent lift increments reverse their ordering between the two orientations. The common-axis moment comparison also changes from reinforcement to compensation between the contributions from F_x and F_z across incidence intervals. Both the force magnitudes and their moment arms enter this incidence-dependent response.

The forward-swept planform gives stronger fixed-surface lift growth, much weaker control loss, positive intermediate-chord growth and an inboard-concentrated aft lift loss over the matched $\alpha = 0 \rightarrow 2°$ interval. Its aft moment increments consequently lack the reference model's increase in positive moment contribution toward the tip about the corresponding aerodynamic centers. This connects the different pitching responses to the chordwise and spanwise organization of the computed loading.

Future computations on the finer levels of the established mesh hierarchy are planned to assess the resolution sensitivity of the pre-stall load redistribution, with particular attention to pitching moment and the fixed- and control-surface contributions. The present joint analysis identifies how continued lift growth can coexist with different component lift deficits and pitching-moment contributions.

A. Sensitivity Checks

This appendix checks three choices made in the analysis. The departures depend on the preceding incidence samples that define each fitted trend (Sec. II.D). Their regional values also depend on how the surface is divided (Sec. IV.B). The reverse-flow area fraction depends on the sign criterion $C_{f,c} < 0$ (Sec. II.D). The checks follow this order, from the fitted trend to the regions and then to the near-wall flow. The appendix ends with two features of the stored solutions that are not interpreted.

The first check concerns the preceding trend. Table 7 lists the reference $\delta = 20°$, $\alpha = 0 \rightarrow 2°$ lift allocation for three preceding-incidence fitting ranges. The fixed-surface fraction remains 62.4–62.6%, while the total lift departure varies from approximately −0.105 to −0.137. The allocation therefore varies much less than the departure amplitude. These rows quantify robustness to the empirical preceding trend; they are neither a confidence interval nor a grid-convergence estimate.

Table 7 Reference loading departures over $\alpha = 0 \rightarrow 2°$, $\delta = 20°$: sensitivity to the preceding incidence samples; moments are about the computed aerodynamic center.

Preceding α (deg)	Total D_L	Fixed fraction (%)	Total D_M
$\{-6, -4, -2\}$	-0.1106	62.55	+0.0332
$\{-8, -6, -4\}$	-0.1368	62.43	+0.0409
$\{-4, -2, 0\}$	-0.1050	62.64	+0.0340

With the complete departures checked, the same question applies to their division into regions. Figure 19 shows the two partitions used to check the regional interpretation. The first divides the semispan into thirds and the chord at the hinge; the second uses spanwise quarters and adds a chord boundary at $\xi = 0.4$.

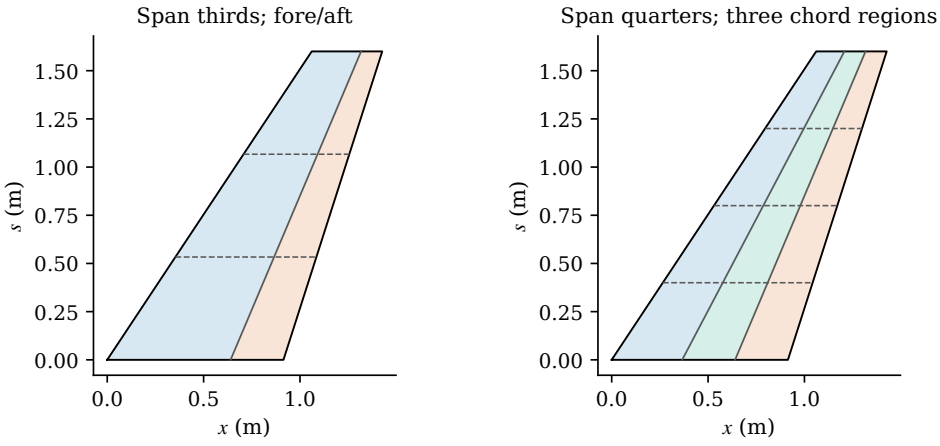


Fig. 19 Alternative integration partitions: spanwise thirds with fore/aft regions (left), and quarters with forward/intermediate/aft regions (right).

Moving the spanwise boundary from one third to one quarter changes the inboard aft departure from approximately

−0.0127 to −0.00939, while complete fore/aft sums remain unchanged. Finer chord subdivisions preserve the two negative fixed-chord departures. These checks support the complete loading interpretation while showing that individual regional contributions depend on the partition.

The regional loads are compared in the main text with the near-wall flow on the control, measured by the reverse-flow area fraction. Replacing the zero threshold with small negative values, $C_{f,c} < -10^{-6}$ and $C_{f,c} < -10^{-5}$, leaves the reference result nearly unchanged. With the larger threshold, the $\delta = 20°$ fraction grows from 0.145 to 0.480 across $\alpha = 0 \rightarrow 2°$, compared with 0.146 to 0.485 in Fig. 9. These threshold checks and the fractions over other geometric partitions are retained in the reproducibility record; the main figures use $C_{f,c} < 0$.

The same measure shows the first of two features in the stored $\Gamma = 45°$ solutions that are not interpreted. The upper control surface reverse-flow area fractions decrease between $\alpha = 2°$ and $\alpha = 4°$, from about 0.58 to 0.09 at $\delta = 20°$ and from 0.91 to 0.20 at $\delta = 30°$. Their physical origin is unresolved, so they are not interpreted as a transition or reattachment sequence. The second feature is in the integrated loads. The $\delta = 20°$ lift irregularity over $\alpha = -6 \rightarrow -4°$ exceeds the coefficient variations in the final iterations, of order 10^{-4}, but this does not establish a unique steady state.

Acknowledgments

This project has received funding from the Clean Sky 2 Joint Undertaking (JU) under grant agreement No 101008257. The JU receives support from the European Union's Horizon 2020 research and innovation programme and the Clean Sky 2 JU members other than the Union.

References

[1] Sears, R. I., and Liddell, R. B., "Wind-Tunnel Investigation of Control-Surface Characteristics VI: A 30-Percent-Chord Plain Flap on the NACA 0015 Airfoil," Tech. Rep. NACA-WR-L-454, National Advisory Committee for Aeronautics, 1942. URL https://ntrs.nasa.gov/citations/19930092872.

[2] Campbell, I. J., Blanks, C. F., and Leaver, D. A., "Aerodynamic Characteristics of Rectangular Wings of Small Aspect Ratio," Tech. Rep. Reports and Memoranda No. 3142, Aeronautical Research Council, 1956. URL https://reports.aerade.cranfield.ac.uk/handle/1826.2/3709.

[3] Harper, C. W., and Maki, R. L., "A Review of the Stall Characteristics of Swept Wings," Tech. Rep. NASA TN D-2373, National Aeronautics and Space Administration, 1964. URL https://ntrs.nasa.gov/citations/19640014908.

[4] Tewes, P., Taubert, L., and Wygnanski, I. J., "On the Effect of Sweep on Separation Control," *7th AIAA Flow Control Conference*, Atlanta, GA, 2014. https://doi.org/10.2514/6.2014-2513.

[5] Melton, L. P., "Active Flow Separation Control on a NACA 0015 Wing Using Fluidic Actuators," *7th AIAA Flow Control Conference*, Atlanta, GA, 2014. https://doi.org/10.2514/6.2014-2364.

[6] Sivells, J. C., and Neely, R. H., "Method for Calculating Wing Characteristics by Lifting-Line Theory Using Nonlinear Section Lift Data," Tech. Rep. NACA Report 865, National Advisory Committee for Aeronautics, 1947. URL https://ntrs.nasa.gov/citations/19930091938.

[7] Goitia, H., and Llamas, R., "Nonlinear Vortex Lattice Method for Stall Prediction," *MATEC Web of Conferences*, Vol. 304, 2019, p. 02006. https://doi.org/10.1051/matecconf/201930402006.

[8] Rausa, A., Gori, G., Zanotti, A., Gibertini, G., Guardone, A., and Auteri, F., "Multifidelity Numerical–Experimental Database for Tailplane Aerodynamics," *AIAA Journal*, 2026, pp. 1–12. https://doi.org/10.2514/1.J066373, article in Advance.

[9] Tugnoli, M., Montagnani, D., Syal, M., Droandi, G., and Zanotti, A., "Mid-Fidelity Approach to Aerodynamic Simulations of Unconventional VTOL Aircraft Configurations," *Aerospace Science and Technology*, Vol. 115, 2021, p. 106804. https://doi.org/10.1016/j.ast.2021.106804.

[10] Palacios, F., Alonso, J., Duraisamy, K., Colonno, M., Hicken, J., Aranake, A., Campos, A., Copeland, S., Economon, T., Lonkar, A., Lukaczyk, T., and Taylor, T., "Stanford University Unstructured SU2: An open-source integrated computational environment for multi-physics simulation and design," Tech. Rep. AIAA Paper 2013-287, American Institute of Aeronautics and Astronautics, 2013. https://doi.org/10.2514/6.2013-287.

[11] Spalart, P. R., and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," Tech. Rep. AIAA Paper 1992-439, American Institute of Aeronautics and Astronautics, 1992. https://doi.org/10.2514/6.1992-439.

[12] Cakmakcioglu, S. C., Bas, O., Mura, R., and Kaynak, U., "A Revised One-Equation Transitional Model for External Aerodynamics," Tech. Rep. AIAA Paper 2020-2706, American Institute of Aeronautics and Astronautics, 2020. https://doi.org/10.2514/6.2020-2706.

[13] Van Leer, B., "Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method," *Journal of computational Physics*, Vol. 32, No. 1, 1979, pp. 101–136.

[14] Venkatakrishnan, V., "On the accuracy of limiters and convergence to steady state solutions," Tech. Rep. AIAA Paper 1993-880, American Institute of Aeronautics and Astronautics, 1993. https://doi.org/10.2514/6.1993-880.

[15] Tobak, M., and Peake, D. J., "Topology of Three-Dimensional Separated Flows," *Annual Review of Fluid Mechanics*, Vol. 14, 1982, pp. 61–85. https://doi.org/10.1146/annurev.fl.14.010182.000425.

[16] McCormack, G. M., and Cook, W. L., "A Study of Stall Phenomena on a 45 Degree Swept-Forward Wing," Tech. Rep. NACA TN 1797, National Advisory Committee for Aeronautics, 1949. URL https://ntrs.nasa.gov/citations/19930082472.

[17] Maggin, B., and Shanks, R. E., "The Effect of Geometric Dihedral on the Aerodynamic Characteristics of a 40 Degree Swept-Back Wing of Aspect Ratio 3," Tech. Rep. NACA TN 1169, National Advisory Committee for Aeronautics, 1946. URL https://ntrs.nasa.gov/citations/19930081920.

[18] Toppings, C. E., and Yarusevych, S., "Structure and Dynamics of a Laminar Separation Bubble near a Wingtip," *Journal of Fluid Mechanics*, Vol. 929, 2021, p. A39. https://doi.org/10.1017/jfm.2021.881.